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Innovative Digitalization of Train Maintenance

A Mapping of Digital Tools and Technologies to Facilitate Technicians Performing Maintenance

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Innovativ digitalisering av tågunderhåll

En kartläggning av digitala verktyg och tekniker för att underlätta för fordonstekniker som utför underhåll

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Sammanfattning

I takt med att digitala verktyg och tekniker såsom AR, VR, Digital Tvilling, IoT och andra interaktiva visualiseringsteknologier utvecklas uppstår nya möjligheter att använda dessa för att optimera effektiviteten inom verksamheter, inte minst inom området för underhåll. En bransch där underhåll är av största vikt är järnvägsbranschen, där målet att ha en punktlig, pålitlig och kostnadseffektiv järnvägstrafik är beroende av tillgången av underhållna spårfordon. Underhåll på fordon utförs av fordonstekniker som arbetar i depåer för att åtgärda fel och serva fordonen. Brist på resurser, dålig samordning och informationshantering samt komplexa fordon gör dock fordonsteknikernas arbete och rutiner besvärliga.

Syftet med detta examensarbete var att undersöka och ge rekommendationer hur digitala verktyg och tekniker kan underlätta underhållet av spårfordon. För att uppnå det har en kvalitativ studie genomförts bestående av 12 intervjuer med fordonstekniker och depåpersonal. Syftet med studien var att förstå de största utmaningarna och tidskrävande aktiviteter de stöter på i sitt arbete. Vidare har potentiella användningsområden hos digitala verktyg och tekniker för att underlätta de identifierade utmaningarna analyserats för att fastställa rekommendationer om vilka verktyg och tekniker som kan användas och hur de bör användas för att hjälpa fordonsteknikerna. De främsta utmaningarna och tidskrävande aktiviteterna som hittats inkluderar att snabbt och enkelt hitta och komma åt relevant information, förstå tillgänglig information, samt samarbetet med aktörer för stöd och expertis. De digitala verktygen och teknikerna som rekommenderas består av surfplattor för att nå information snabbt och enkelt, AR för att visualisera instruktioner och skapa en miljö för fjärrsupport, AI för att snabbare söka och filtrera användbar information för att effektivisera felsökningsprocedurerna, Digital Modell för att visualisera fordonskonfiguration samt länka och strukturera information, och slutligen eMaintenance teknik för överföring av realtidsdata för att snabbare få information om fordonet och dess fel. Resultatet av detta examensarbete kan ge en riktning för framtida forskning, utveckling och integration av digitala verktyg och tekniker för att underlätta för fordonstekniker i underhållet av spårfordon.

Nyckelord: Underhåll, Fordonstekniker, Spårfordon, Digitala Verktyg och Tekniker, AR, VR, Digital Tvilling, Digital Modell, AI, Visualisering, eMaintenance

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Abstract

As digital tools and technologies, such as AR, VR, Digital Twin, IoT and other interactive visualization technologies are being developed, new opportunities arise to use these to optimize the efficiency of operations, not least in the area of maintenance. One industry where maintenance is of utmost importance is the railway industry, where the goal of having a punctual, reliable and cost-effective railway traffic is dependent on the availability of maintained railway vehicles. The maintenance procedures of the vehicles are performed by technicians, working in depots to fix faults and service the vehicles. However, a lack of resources, poor coordination and information handling, and complex vehicles make the work and procedures of technicians cumbersome.

The purpose of this thesis project was to investigate and give recommendations on how digital tools and technologies can facilitate the maintenance of railway vehicles. To do so, a qualitative study has been performed consisting of 12 interviews with technicians and depot staff. The purpose of the study was to understand their main challenges and time-consuming activities. Moreover, the potential use of digital tools and technologies to facilitate the identified challenges has been analyzed to establish recommendations on what tools and technologies can be used and how they should be used to aid technicians. The main challenges and time-consuming activities found include finding and accessing relevant information quickly and easily, understanding the information available, and collaborating with actors for support and expertise. The digital tools and technologies recommended consist of tablets to access information quick and easy, AR to visualize instructions and create an environment for remote support, AI to quickly search and filter information to enhance troubleshooting procedures, Digital Model to visualize vehicle configuration and link and structure information, and lastly, real-time data transfer utilizing eMaintenance technologies to faster receive information about the vehicle and its faults. The results of this thesis project can give direction for future research and the development and integration of digital tools and technologies to facilitate technicians in the maintenance of railway vehicles.

Keywords: Maintenance, Technician, Railway Vehicles, Digital Tools and Technologies, AR, VR, Digital Twin, Digital Model, AI, Visualization, eMaintenance

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List of Abbreviations

AI - Artificial Intelligence

AR - Augmented Reality

BIM - Building Information Model

CAD - Computer Aided Design

CM - Corrective Maintenance

HHD - Hand-Held Display

HMD - Head-Mounted Display

HVAC - Heating, Ventilation and Air Conditioning

ICT - Information and Communication Technology

IoT - Internet of Things

MR - Mixed Reality

PM - Preventive Maintenance

VR - Virtual Reality

XR - Extended Reality

1 Introduction

This chapter consists of the background and introduction to the subject researched and studied in this thesis project. Furthermore, the problem description, purpose and research questions, and delimitations are presented.

1.1 Background

The development and implementation of new technology to enhance operational efficiency within industries has been prominent in recent years, often referred to as the fourth industrial revolution called Industry 4.0 [1]. Industry 4.0 is distinguished by incorporating internet-based solutions, cyber-physical systems [2] and smart devices [3] into industrial operations. The integration of advanced technologies such as Internet of Things (IoT), Artificial Intelligence (AI), Digital Twins, Big Data, Extended Reality (XR) including Augmented Reality (AR), Mixed Reality (MR) and Virtual Reality (VR), has been part of the shift in industry [4]. This shift is more technologically advanced compared to traditional methods such as lean practices and the use of 5S to achieve higher efficiency [5] by creating a well-organized and clean workplace by sorting away unnecessary things (sort), creating a logical order of items (set in order), maintaining a clean workspace (shine), having rules for storage to avoid mistakes and search times (standardize), and evaluation of methods to improve the workspace (sustain) [6]. One particular area where opportunities to use digital tools and technologies are researched is the area of maintenance [7],[8]. Studies show that utilizing digital tools and technologies can improve the efficiency and quality of maintenance processes.

The shift towards Industry 4.0 has also been seen in the railway industry as railway vehicles manufactured today are smarter and more connected than ever. Possibilities of having integrated systems and sensors on trains to collect and report large amounts of data are present and can be used to support operational decisions in terms of design and asset management [9],[10]. However, less research has been devoted to utilizing these digital tools and technologies to facilitate the work and procedures of technicians performing maintenance on railway vehicles.

1.2 Problem Description

The deregulation of the Swedish railway system, which started in 1988 and ended in 2011 [9], has characterized the railway industry of today. This deregulation resulted in several different actors within the railway ecosystem, both concerning infrastructure as well as rolling stock, including vehicles such as trains or trams. These actors can be part of either one or several activities and processes, making the system rather complex [2], [9]. Maintenance is a big part of this ecosystem, and it is vital for having a well-functioning railway system. However, due to aging infrastructure and rolling stock combined with high capacity utilization of the Swedish railway, it is hard to fulfill the maintenance demands [9]. Furthermore, a lack of resources together with poor coordination, communication and information handling between actors makes the maintenance process challenging. The goal for the Swedish railway traffic is for it to be punctual, reliable and cost-efficient [11], and to reach this goal is the maintenance of railway vehicles essential [12]. Whether it is corrective maintenance, performed after a failure has occurred, or preventive maintenance activities, both are relevant for ensuring that railway vehicles can be put back into traffic quickly and safely. In terms of the maintenance process of rolling stock, several actors are involved and between these actors are different contracts and collaborations present as well as an exchange of data, information and documentation of different sorts. The complexity is therefore present in terms of the roles and responsibilities of each actor as well as documentation and information sharing and handling.

Furthermore, railway vehicles are complex vehicles with several integrated systems [13]. The vehicles are not only made up of mechanical systems, but also mechatronic systems combining mechanical, electrical and information technology systems [14]. Maintenance procedures such as troubleshooting are important to solve the problems and faults present in the vehicle. However, it can be difficult for technicians to perform as different data, documentation and information from multiple sources and with different structures are used. Furthermore, the limitation of IT knowledge as well as a lack of fruitful collaboration among actors results in knowledge gaps when trying to determine failures of the vehicle [13]. In order for technicians to perform efficient and high-quality maintenance with ease, new innovation is required [11]. The use and development of innovative digital tools and technologies to optimize maintenance operations are on the rise within

industries, but what opportunities may they bring in the work and procedures of technicians performing maintenance on railway vehicles?

1.3 Purpose and Research Questions

The purpose of this thesis project is to discover ways in which innovative digital tools and technologies can be used to help technicians perform maintenance procedures on railway vehicles. The aim is to establish the maintenance process of railway vehicles, to highlight the challenges and time-consuming activities of the maintenance process for a technician, and to recommend digital tools and technologies that can facilitate the work of technicians and make the maintenance process more efficient. Furthermore, the purpose is to give insights into the applicability of these tools and technologies, how the data and information available can be used with these tools and technologies, and future implementations or research areas. To fulfill the purpose of the thesis project, the following research question has been formulated:

- How can digital tools and technologies facilitate the maintenance of railway vehicles?

To answer the research question in detail, sub-questions have been formulated.

  - What are the main challenging activities for technicians performing maintenance on railway vehicles?

  - How can various digital tools and technologies be used to facilitate technicians performing maintenance on railway vehicles?

1.4 Delimitations

The focus is on maintenance operations and procedures performed by technicians and what digital tools and technologies may facilitate their work. The study does not address maintenance operations on a higher organizational level, such as asset management, planning and scheduling of maintenance, or tools and technologies to optimize predictive maintenance. Furthermore, only Swedish depots performing maintenance on different types of railway vehicles are visited and interviewed, hence the results may only be appropriate to the maintenance process and ecosys-

tem of rolling stock in Sweden. The scope of the thesis includes a mapping of potential digital tools and technologies and a recommendation of digital tools and technology concepts that may facilitate the challenges identified. The scope of the thesis project did not include the development or implementation of the tools and technologies recommended, due to time constraints, the limitation of knowledge and experience by the author, and the exploratory nature of the thesis project.

2 Frame of Reference

This chapter presents the maintenance process and ecosystem of rolling stock as well as a comprehension of research literature within the world of digital technologies used to enhance operational efficiency. Furthermore, relevant research areas, technologies and tools that aim to facilitate maintenance processes and procedures are presented.

2.1 Swedish Railway Rolling Stock Ecosystem

As mentioned in the introduction, the railway ecosystem in Sweden consists of many different actors, both concerning infrastructure as well as rolling stock [9]. The amount of actors and the fact that the roles and responsibilities of each actor can differ depending on the traffic contract [2], makes the ecosystem rather complex [9]. When describing the system from a rolling stock perspective, there are four main functions to consider; the one who manufactures the vehicle, the one who is responsible for the vehicle and manages it, the one who maintains the vehicle and the one who operates the vehicle [9]. Figure 1 presents a simplified overview of relevant actors involved in the operation and maintenance of rolling stock.

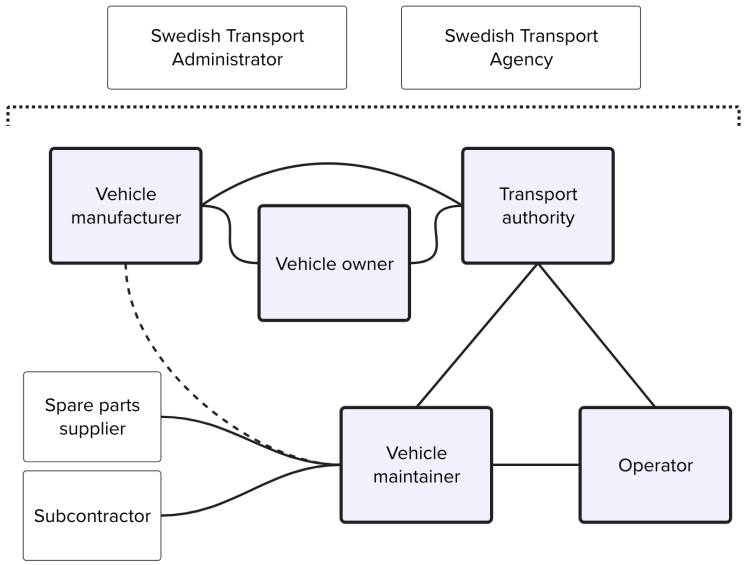


Figure 1: Ecosystem of actors involved in the operation and maintenance of rolling stock. Based on [9], [15].

The transport authority is responsible for the traffic system and procure actors such as the operator, who drives the vehicles and provides transportation services,

and the vehicle maintainer, who performs maintenance on the vehicles. Furthermore, the transport authority is often the actor responsible for managing the vehicles and distributing relevant vehicle information to the vehicle maintainer and the operator. The transport authority can sometimes own the vehicles, but it could also be a separate actor who owns the vehicles [15]. Other actors, who support the vehicle maintainer, are spare parts suppliers and subcontractors. The vehicle manufacturer develops and manufactures the vehicles and supplies relevant technical documents and information to the vehicle manager, in this example the transport authority. They are therefore an actor who sits on important knowledge and information of the vehicle. The vehicle manufacturer can also act as a subcontractor and spare part supplier for the vehicle maintainer. Furthermore, since the vehicles have a warranty period of a couple of years, the vehicle manufacturer is usually in closer contact with the vehicle maintainer at the beginning of the vehicle's lifetime. Other relevant parties shown in Figure 1 are the governmental bodies, the Swedish Transport Administration (Trafikverket) and The Swedish Transport Agency (Transportstyrelsen), actors enforcing decisions, rules and regulations. The Swedish Transport Administration is responsible for the infrastructure and the Swedish Transport Agency is the regulatory body responsible for market surveillance [9], [15].

The flow of information between these actors is described in a case study by Kans et al. [9]. It is mentioned that the vehicle maintainer mainly exchanges information with the actor responsible for managing the vehicle and the actor operating the vehicle. The information consists of the status of the asset, eventual problems related to the asset, and the capability of the maintainer to perform maintenance work. The information regarding the capacity of the vehicle fleet, eg how many vehicles of the fleet can be used in traffic, is transferred between the vehicle manager and the operator for them to plan transportation services. The number of different actors within this flow of information as well as the complexity of extracting the right data or information can hinder the efficiency of the communication. For instance, it is described that the information flow between the vehicle maintainer and the vehicle manufacturer is lacking [9].

2.1.1 Maintenance Regulations of Rolling Stock

The maintenance of railway vehicles in Sweden has since June 2022 been regulated according to the EU regulation 2019/779 called ECM, which stands for Entity in Charge of Maintenance [16]. The purpose of the regulation is to manage vehicles and their maintenance better, ensuring that the entities needed to perform maintenance are established and that the administrative competencies and systems needed are present [12]. The maintenance system, according to the ECM, is composed of four functions: ECM-F1, ECM-F2, ECM-F3 and ECM-F4.

- **ECM-F1** coordinates and supervises the maintenance system and the following ECM-functions and ensures that the railway vehicles are in a safe state.

- **ECM-F2** is responsible for the management of maintenance documentation and configuration management based on the design and operational data.

- **ECM-F3** is responsible for the vehicle's removal when maintenance is due, and the return to operation when maintenance has been performed.

- **ECM-F4** performs technical maintenance tasks on the railway vehicle or parts of it as well as the release of service documentation.

The ECM-1 is the main function that needs to be appointed to an actor for the vehicle to be able to be put into service [12]. This actor is the entity that is responsible for that maintenance can be performed on the vehicle, that the ECM functions are coordinated and that the vehicle is in a safe state [16]. The stakeholder or actor responsible for a certain function differs from each case of traffic contract or vehicle fleet.

2.1.2 Maintenance Strategies of Rolling Stock

Different types of maintenance strategies are performed on railway vehicles. The maintenance strategies, according to the SS-EN 13306:2017 standard, can be divided into different categories [10], [12], Corrective Maintenance, or also referenced to as Reactive Maintenance [17], and Preventive Maintenance. The different maintenance strategies and their divisions are displayed in Figure 2.

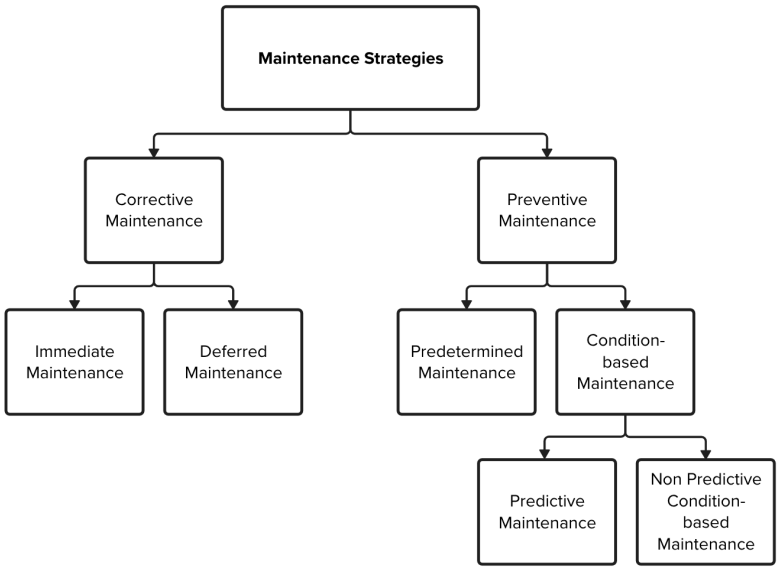


Figure 2: Maintenance strategies performed on railway vehicles. Based on SS-EN 13306:2017, [10], [12], [17].

Corrective Maintenance consists of actions performed when failure has already occurred [12]. The intention is to restore the asset so it can perform its intended function [12], [17]. This type of maintenance is either categorized as immediate or deferred depending on the degree of the failure. It is classified as immediate maintenance if actions need to be taken immediately. It is classified as deferred maintenance if the failure is not that urgent and if it can be deferred to a time when preventive maintenance is planned to be performed. Since the nature of corrective maintenance is rather unpredictable, the maintenance plan for this type of maintenance is hard to establish [12]. This type of maintenance can be performed by either a maintainer or the vehicle manufacturer during the warranty period of the vehicles.

Preventive Maintenance, on the other hand, consists of actions taken to prevent failure. Predetermined maintenance is planned or scheduled in advance according to specific intervals and no condition investigation is performed. Another kind of preventive maintenance is Condition-Based Maintenance in which condition monitoring and assessment are used together with inspection, test, and analysis. Predictive Maintenance is when a prognosis of the degradation is used to take action before faults occur. All Preventive Maintenance of a railway vehicle is governed by the maintenance plan, which is a document including information about what maintenance procedures need to be performed, with what periodicity, and how

they should be performed [12]. The vehicle manufacturer is the actor providing the maintenance plan to ensure that the actors responsible for the vehicle and the actors performing maintenance can uphold vehicle safety during its lifetime.

2.2 Maintenance 4.0

One application area of Industry 4.0 is maintenance, usually referenced to as Maintenance 4.0 [2]. With the help of advanced technologies and smart systems, such as sensors, IoT and different information systems, collected data, such as log files, information about faults, and system operations can be analyzed to maintain assets better and make informed decisions [2],[3], [18]. Maintenance 4.0 is also being discovered within the railway industry and how different technologies can bring benefits to organizations and operations from a system perspective [2],[10].

Examples of supportive technologies utilized to enable Maintenance 4.0 are sensors and Radio-frequency identification (RFID). These technologies are used to monitor assets or systems and their health to guide decisions in maintenance operations. Sensors are used to monitor the condition of assets and enable alarm messages if certain events happen [10], and RFID is a technology used for collecting information from assets from a distance [2]. By using a tag, a reading device together with an antenna, and middleware in the form of computing software, tracking of objects, such as spare parts [19] or vehicles [2] can be achieved. With the utilization of supportive technologies, more transformative technologies can be implemented. These technologies utilize a real-time data-driven approach to ensure higher reliability and quality of operations and maintenance services. Technologies such as AI, machine learning, Big data analytics, IoT, eMaintenance and visualization technologies are examples of technologies that can transform railway maintenance practices, enabling more effective and efficient processes [2], [10]. AI can be used to perform more advanced analytics within asset management with the help of different data sources such as maintenance history, log files, model data information from manufacturers, sensor data and images or videos [10]. Jarrahi et al. highlight the power of AI for knowledge retrieval as it can analyze and filter several sources and information portals simultaneously [20]. The utilization of AI technology can make the process of filtering and finding information more efficient [18]. However, it is established that there needs to be data of high quality and quantity to learn the AI system, putting pressure on data management and

structure [18], [20].

Furthermore, Big Data analytics is on the rise as it becomes more relevant when employing IoT and collecting vast amounts of data [10]. However, it is described that there is a challenge in understanding and structuring the data collected and utilizing it to make better-informed decisions. There will be a need to harmonize and structure the incoming data by developing standards and assessing the data uncertainty to understand and exploit the information gathered [2], [10]. Caporuscio et al. [21] highlight the importance of effective logging systems in order to subtract relevant information. They suggest a centralized logging unit to capture information from different connected devices and then present it in a standard easy-to-use format. EMaintenance, is an example of when big data analytics is used to analyze data and provide decision guidance [10]. It is a broad term describing simple digital administration systems as well as more advanced support systems and technologies providing diagnostic and decision guidance in maintenance [2]. It includes various technologies and activities such as Information and Communication Technologies (ICT) and monitoring, diagnosis and prognosis. It is essentially a combination of information, knowledge and analysis to support decisions [10]. One example where an information system is used to gather real-time data and then later process and analyze the data is the Orbita Fleet Monitor system, OFM, provided by Bombardier [17]. The system provides events of the operation of the vehicle during traffic as well as the circumstances or environment present when the events happened, such as the speed of the vehicle and its location. It is described that the utilization of OFM can serve as beneficial as it can support the troubleshooting process and the determination of the root cause of a fault. Furthermore, it is described that the utilization of this system can shorten the time spent on corrective maintenance and also heighten the ability to better plan for condition-based maintenance and optimize the maintenance intervals of scheduled maintenance.

2.3 Digital Twin, Digital Shadow and Digital Model

There are many different definitions of a digital twin. Coupry et al. [8] describe that the purpose of a digital twin is to use real-time collected data together with historical data to perform simulations using algorithms to predict the future state of the asset and optimize the operation. Kumar et al. [10] describe it as *"a dy-*

namic digital model of a system or a physical asset". The phrase digital twin has been used while describing different levels of digital integration [8], [22], [23]. The different levels of integration describe the manual or automatic flow of data between the digital and the physical asset and it is shown in Figure 3. In this thesis, the definition of digital twin, where it is described in relation to two other levels of manual or automatic flow of data between a physical and a digital asset, has been used. The first level is the so called Digital Model, which is a digital representation of the physical asset, but there is no automated data transfer. The Digital Model can therefore be considered as a 3D information model of the asset which can have information and data linked to it [8]. The second level is the Digital Shadow, which is a digital representation of the physical asset where data flows automatic from the physical asset to the digital asset. Lastly is the Digital Twin, the digital representation where data flows automatic between both the physical and the digital asset. In this way, one can control the actions of the physical asset while having access to the digital asset.

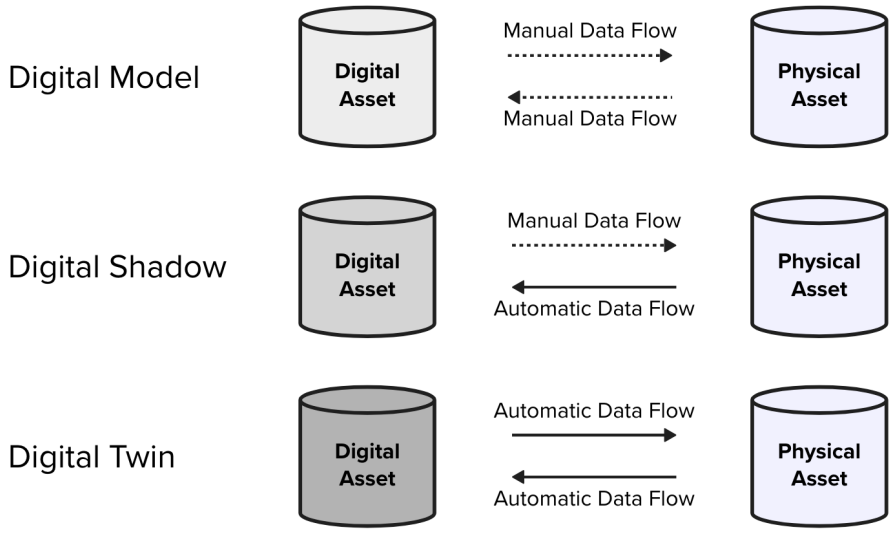


Figure 3: Integration levels of digital representations and the flow of data between the physical asset and the digital asset. Based on [8], [22], [23].

The application of a digital twin varies. Kumar et al. [10] describe how a digital twin can be used to connect complex assets, such as a train, and their operational technology into a digital environment. The digital twin can become a virtual representation of a physical asset and its systems, which can be used to monitor its performance. It is however emphasized that the digital representation is not

complete without the engineering knowledge, which is not typically used during the operational phase. Therefore, it is important to include engineering knowledge, such as information from the design and manufacturing, into a digital twin in order to have relevant data about the physicality of the asset to complete the information model. The use of a digital twin can also serve as beneficial when maintenance costs are high and there are difficulties in accessing the physical asset [23].

Other studies point to the use of a digital twin to improve the monitoring and planning of the operation and maintenance of an asset [8]. Coupry et al. [8] present the way of using a Building Information Model (BIM) as a basis for the digital twin linked to a centralized database. In this way, real-time data, collected from an asset and static data such as technical documentation and maintenance history, can be presented and linked to a 3D representation of an asset. With the help of installed sensors and prediction analysis, potential faults and deterioration of components can be detected and maintenance strategies can become more predictive rather than corrective. However, the data processing part of this technology is important and needed to gather data from sensors, process them using algorithms, and also exchange data and information between the physical and digital assets. Other challenges are implementation, selection of data, lack of standards in terms of data, and new technical skills required within the organization.

2.4 Extended Reality

Extended reality, XR for short, is a term encompassing different environments created using immersive technologies such as AR, MR and VR, ranging from the least to the most immersive [8]. The use of these technologies is found in different industries and for different application purposes such as manufacturing, design, maintenance and marketing [8], [10], [24].

2.4.1 Augmented Reality

AR is a technology used to place virtual objects or information onto the real world by using different kinds of equipment [4]. This technology can be used to enable information at the appropriate time during maintenance procedures [10], [13], [25] while transforming manuals existing on paper into electronic media that can be accessed hands-free [26]. Furthermore, one advantage mentioned is the ability to provide additional context information that can be immediately interpreted by

users [26]. An AR system is built with different components, which usually are a visualization device, a capturing device, an interaction device as well as a tracking system [13]. The capturing technologies collect data from the surrounding environment while the visualization part of the system is used to display information [13], either with a hand-held display (HHD), like a mobile device or a tablet, a head-mounted display (HMD), like smart goggles, or a spatial display, like a projector [27]. The interaction part of the system consists of a device that can take in commands and adapt the information processing and what is being displayed [13]. Lastly, the tracking part of the system is used to identify the position of the user to provide correct augmented information. The two most common tracking systems used are marker-based tracking and marker-less tracking [13], [28]. Marker-based tracking uses physical tags on elements for the AR system to recognize. Marker-less tracking systems, feature-based or model-based, can with the help of a thorough and accurate data structure, in the form of for example computer-aided design models (CAD) or 3D point clouds, use the environment's features instead of physical markers, eliminating the need for object tracking.

The field of application of AR in maintenance operations is broad, a literature review has shown that industries such as aviation, automotive, railway, and military are examples of industries where this application is relevant [28]. Furthermore, common maintenance operations mentioned that could benefit from implementing AR are assembly, repair, inspection and diagnosis, and lastly training. When developing an AR system there are usually one or several development platforms utilized. The most used development platforms are based on language programming. The most used programming languages are; C++, C#, java, HTML, CSS, Python etc. Libraries of functions are also used as development platforms, such as OpenCV or OpenGL. These two types of development platforms ensure high flexibility since the system development is performed from scratch, but the skill level required to develop such a system is rather high [28]. Other development platforms mentioned are Game Engines. Game engines are usually more user-friendly when building the application since they do not require solid knowledge of programming languages. The most common game engines used are Unity3D and Unreal. The most common visualization methods used to overlay computer-generated information onto the real world are dynamic and static 2D/3D information. Dynamic 2D/3D visualization includes animations to give more vivid instructions and is considered more effective than paper instructions. Static 2D/3D visualization in-

cludes static models with information. Other ways to overlay information are text, displayed in labels and with highlighting, or audio.

There are numerous benefits to implementing AR technology in maintenance practices and procedures according to authors [4], [10], [28]. These benefits include reduction of time spent on a maintenance task, improved quality of maintenance performed and reduction of risks of accidents with remote assistance during maintenance [4]. Since it is a technology that can integrate information into the real world in an immersive manner, user can be given information and knowledge paperless and still keep their attention on the real environment and the task at hand [10]. Cifone et al. [7] highlight in their case study how AR technology can lower the skill level needed in order to perform a task. However, it is discussed that adding digital technology may create a barrier for the technicians to engage with the process and utilize creative thinking. Furthermore, it is emphasized that for an AR system to be applied and used in an easy and efficient manner in maintenance, there is a need for the system to be modifiable as well as adaptive [28]. Potential applications of AR within the railway industry are presented by Scheffer et al. [13]. Results from the qualitative field study point to the fact that AR applications have the best potential when used in maintenance operations, and more specifically for providing task and work description, fault detection, inspection of components, remote control as well as support and assistance in special work cases. It is stated that applying AR in maintenance operations can help the understanding of the failures present. However, it is stated that the solution developed needs to be user-friendly and easy to adapt and learn by the technician for it to be a successful implementation of the technology. In terms of the implementation of AR in maintenance operations, authors Lorenz et al. [29] emphasize the importance of having a solution that is user-friendly, portable and aids with more visual context such as 3D objects or animations. Technical requirements mentioned are data connection regardless of location in the workspace and the importance of connectivity to systems needed such as data storage systems for documentation. Furthermore, AR hardware is discussed, and it is emphasized that AR goggles need to be easy and safe to use in consideration of the work attire and the environment of the workspace. Moreover, it is explained that current AR goggle solutions can lack compatibility for such an environment.

2.4.2 Mixed Reality

Mixed reality, also called augmented virtuality, is when both virtual and real are mixed and where interaction with the physical and digital environment is possible [24], by for example being able to get information from how the user interacts in the real world, and at the same time be able to take control of physical objects through the digital device [8]. This technology has been used for visualization purposes and also for visualizing picking procedures, however, it is the least developed XR technology on the market [24].

2.4.3 Virtual Reality

Virtual Reality is a more immersive technology in which the user is fully immersed in the virtual world or environment [24]. By using similar development systems and platforms as AR technology can the virtual world be built in 3D and users can access it by wearing HMDs and hand controllers. The user can move and interact in the virtual world, without seeing the real world. VR technology has applicability in different industries and is often used for application within product development, training and visualization.

Kumar et al. [10] describe how VR can be used within maintenance to train and educate personnel. The technician can immerse themselves into a virtual working environment to explore, learn and practice activities without the risk of damaging the physical asset. Nikitin et al. [30] present an example of this in their article. They develop and test a prototype simulator based on VR to evaluate the skills of rolling stock inspectors and their ability to follow instructions. It was designed on the HTC Vive virtual reality system with a compatible PC with VR tools such as Blender, 3ds Max, Unity, Steam VR and C#. The simulation had two users incorporated, the inspector; the person working in the VR setting who should detect faults and report them and, the instructor; who uses the PC monitor screen to assign faults and monitor the work of the inspector. By utilizing this simulation, maintenance personnel could be prepared with experience and skills before performing the actual maintenance tasks, but also be retrained if needed.

2.5 Smart Documents

Digital tools and technologies to visualize schematics and other documents are studied to optimize maintenance operations [31], [32]. One example is the technical diagram visualization system, SDViz, which is a tool that maintenance technicians can use to explore and navigate technical diagrams to more easily diagnose faults [31]. By enabling contextual information between diagrams, such as highlighting, spatial information and working status together with search functions within and across diagrams and documents, the maintenance operations performed by technicians can be facilitated, ultimately helping them perform tasks and troubleshoot in a more efficient manner. The function of highlighting components and their connections within a technical diagram and preserving that information between technical diagrams was evaluated to be the most important technique from the field study conducted, as it would guide the user [31]. Other studies show benefits in adding 3D information to components in interactive and visualized schematics to easily understand connections between components and their actual placement [32].

2.6 Examples From Other Industries

The digital tools and technologies mentioned have also been researched and tested in other industries to facilitate maintenance work, procedures and processes. Below are various examples from different industries presented.

2.6.1 Manufacturing Industry

One example of research about tools and technologies to facilitate maintenance within the manufacturing industry is the study performed by Bottani et al. [33], where connected machines via IoT was linked together with two solutions, both a mobile application developed in Java for Android mobile devices and an MR application solution together with a Hololens developed using Unity game engine, C# and Blender to create 3D models and environments. It was developed to provide personnel working in an aseptic bottling line with alerts of machine faults and instructions on how to solve the problem. The two different solutions were tested on operators, technicians and responsible personnel. With grading using the System Usability Scale, [34], it was established that the mobile app was the preferred solution as it was more user-friendly, quick to use, and easy to handle

compared to the Hololens solution. However, it was described that potential was seen in using the Hololens solution while dealing with complex interventions as it allowed for hands-free handling.

2.6.2 Aviation Industry

Another industry where studies about the use of digital tools and technologies have been carried out is the aviation industry. One example is the case study by Rios et al. [35]. The purpose was to develop a solution that could aid the technician during training in troubleshooting skills as well as help with tips from experienced operators while performing maintenance tasks on a 737 Engine Bleed Air system. A solution utilizing AR consisting of a webcam and computer screen was developed and tested for an assembly process. Results showed that the use of AR instructions, compared to traditionally written instructions, decreased the average time by 17.22% and increased the quality of the assembly by 24%. The AR system also allowed for mail communication with important stakeholders and access to up-to-date documentation and manuals. However, the type of assembly performed was clearly instructed with step-by-step information, guiding the technicians step-wise.

Scholars Bergkvist et al. [36] investigate smart solutions for aircraft maintenance. They compare different digital tools and technologies to facilitate the maintenance of an aircraft such as digital twin, XR technologies, tablet applications and voice guidance. Requirements of paperless, traceability and authorization of tasks were established to be the basis of the prototype development. The technology chosen to be the most suitable for maintenance operations was a tablet application due to its maturity and ability to meet the requirements. The demonstrator developed consisted of a tablet application solution using Novacura Flow Studio. This resulted in the solution having features such as user-friendly attributes of both software and hardware, efficient data handling, and logging functions providing the maintenance staff with a better work procedure.

Another example of the use of AR is the case study done by De Crescenzio et al. [37]. They developed a prototype of an AR system to both train and help technicians in their maintenance work of an aircraft. The specific case concerned the development of an AR system to give guidance while performing an engine oil level check. The AR system displayed information with the help of digital

replicas of parts together with symbols of arrows and pointers layered on top of the real world using marker-less camera pose estimation and a prototyped HMD with a camera. Participants tested the system on a step-by-step maintenance procedure and results showed that the efficiency of the operations increased and that participants were satisfied with the visual elements of the solution, such as animation of parts.

2.6.3 Facility Management Industry

Within facility management, and more specifically the maintenance of HVACs, authors Yang et al. [32] describe how the troubleshooting process of HVACs could be more efficiently performed while having required information visually presented. The study compares different types of visualization methods such as 2D or 3D models together with different ways of displaying additional information, such as the location of the HVAC system, control relationship within the system, component dynamic information, and component historical information. The HVAC mechanics were tested on their ability to identify the cause of failure with three different scenarios. Scenario 1 was with information via documents, the accustomed standard approach. Scenario 2 and Scenario 3 utilized the prototype created, displaying information regarding location, control relationships, dynamic information and historical information. The difference was that Scenario 2 only displayed the information in text format, whereas Scenario 3 displayed it using visualizations. The result showed that both Scenario 2 and 3 were performed significantly faster by the user than the traditional way, scenario 1, 35% and 58% in median respectively.

Xie et al. [38] give examples of when AR technologies are used to find faulty components in a building using information data from a BIM-based digital twin with centralized data [38]. By firstly using sensors and analysis can anomalies and faults be detected and alerted to facility managers. By then having access to the location of components in a building from the BIM-based digital twin can AR tools and technologies, such as an HMD, be used to highlight the failed component and its related information with the help of a color-coded 3D model and data to guide operators and maintenance workers to the rightful location. However, it is mentioned that to be able to detect anomalies is a period of monitoring and data retrieval required in order to have enough data to train a model. Furthermore,

the position of both the component or asset as well as the position of the operator needs to be known for this system to work.

2.7 Summary of Digital Tools and Technologies

This section presents a summary of the digital tools and technologies mentioned in the literature and how they can be used, see Table 1.

Table 1: Summary of Digital Tools and Technologies.

Digital Tools and Technologies	**Description**
RFID	Technology used for collecting information from assets from a distance. Examples include the location of an asset.
AI	Technology used for advanced analytics and efficient data filtering.
eMaintenance	Real-time data transfer from an asset to an information system using data analytics and ICT to monitor and analyze asset status.
Digital Twin	Automatic data transfer between a physical asset and digital asset using ICT. With a digital twin, one can have the ability to control the physical asset with the digital asset. Additional information can be linked to the digital asset.
Digital Shadow	Automatic data transfer from a physical asset to a digital asset using ICT. Additional information can be linked to the digital asset.
Digital Model	A digital asset representing a physical asset where data transfer between them is handled manually. Additional information can be linked to the digital asset.
AR	Immersive technology that enables placement of virtual objects onto the real world. This technology can be used for displaying instructions, visualizing the placement of components, or enabling remote support, using tablets or AR goggles.
VR	Technology where a user can fully immersive themselves into a virtual world to see and interact with objects. This technology can be used for training and education as well as for an immersive visualization experience using VR goggles.
Smart documents	Technology to visualize documents digitally and the ability to interact with them. Functions such as highlighting and linking additional information can be used to help guide users.

3 Method

This chapter describes the research design, approach and nature that has been used for this thesis project. Moreover, the specific interview study performed and the thesis process is described in detail to give a thorough background to the qualitative study and the process used for the thesis project. Lastly, the data collection and data analysis methods used are presented.

3.1 Research Design and Approach

The method chosen for this thesis project is a qualitative research method consisting of a review of literature and a situation study in the form of interviews. The qualitative design was chosen to gain deep insights into the maintenance processes performed at rolling stock depots in Sweden and to understand the work and procedures of technicians. The research nature of the thesis is exploratory, aiming to discover potential tools and technologies that could facilitate the work of technicians performing maintenance on railway vehicles. The research had an abductive approach as prior theory gained from literature as well as empirical data from interviews was combined iterative [39]. This approach served as beneficial since it made the research of the subject flexible and findings along the project process could be iterative incorporated.

3.1.1 Situation Study of Rolling Stock Depots

This thesis project was commissioned by WSP Rail Advisory and Twindej, two companies active in the Swedish railway industry. The situation study consisted of field trips to rolling stock depots and interviews. The respondents included technicians and staff from two specific depots and consulting technicians, all performing maintenance on different railway vehicles. The number of interviews and site visits was limited due to time constraints and the availability of technicians and depots to participate. Furthermore, the depots studied and the technicians interviewed were based in Sweden. The results are therefore limited to a perspective based on the structure and practices of rolling stock maintenance performed in Sweden. The initial contact with the depots and respondents in question went via WSP Rail Advisory or Twindej. The situation study was assumed to give a rightful understanding of the maintenance work and procedures performed by technicians and the challenges and time-consuming activities present. It was considered that

findings combined with previous literature would give insights into what tools and technologies could be beneficial to implement within the maintenance operations and how to utilize these to facilitate the maintenance work and procedures of technicians, working on different types of railway vehicles.

3.1.2 Thesis Process

The research method and the process of the thesis project consisted of three phases, see Figure 4.

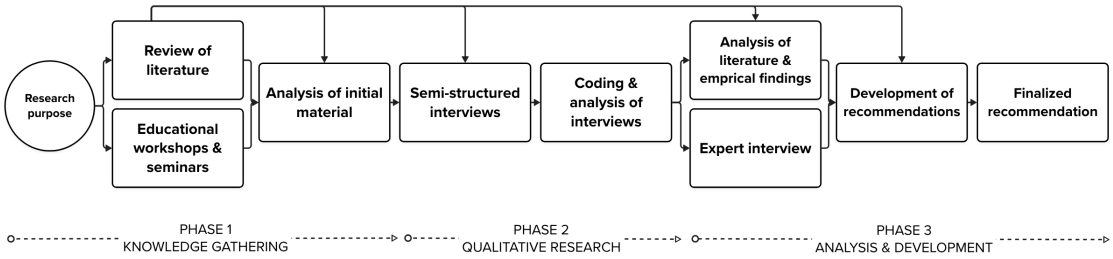


Figure 4: Thesis process.

The first phase, *Knowledge Gathering*, consisted of initial literature research and review together with educational workshops and seminars. The workshops and seminars were three in total and covered areas such as Railway Industry, Train architecture & systems, and digital technologies. The workshops and seminars were held by WSP Rail Advisory and Twindej, together with knowledgeable people within relevant areas. The purpose was to gain deeper insights and knowledge into relevant fields and to get a base for further data collection. The initial material gathered was then analyzed to serve as a direction and basis for the second phase. The second phase, *Qualitative Research*, consisted of multiple semi-structured interviews with maintenance technicians, depot managers and workers to get insights into how maintenance work and procedures are performed and what main challenges and time-consuming activities are present. The interviews were documented and later coded for further analysis to establish common themes. In the third and last phase, *Analysis & Development*, findings from the interviews were analyzed together with the literature. Furthermore, additional research and one interview with a Digital Solution Expert were conducted to get feedback on the applicability of the tools and technologies discussed and analyzed. This served as a basis for the recommendations of tools and technology concepts to facilitate the work of technicians, which were iterated until the final recommendation was established.

3.2 Data Collection

The data collection consists of both primary and secondary sources. The former was collected via semi-structured interviews and the latter through a literature gathering presented as the frame of reference.

3.2.1 Literature

In order to gain a comprehensive overview of the subject area, literature was collected and reviewed. The aim was to get prior knowledge into relevant areas for this thesis project, such as digital tools and technologies, maintenance procedures as well as examples of tools and technologies utilized in other industries. The literature used consists of articles, reports, case studies and conference articles from different academic journals. The sources were collected through databases such as Web of Science, Scopus, IEEE Xplore, and Google Scholar. The keywords used were; Maintenance, Railway, Rolling Stock, troubleshoot, technician, Digital twin, visualization and AR/VR. A manual filtering of sources was conducted to include literature that was of relevance in terms of publication date and scientific journal. Since the research approach chosen for the thesis project was of an abductive character, the frame of reference was built iterative and in parallel with the interviews performed. The choice of digital tools and technologies studied was based on tools and technologies mentioned in related work and technologies discussed during interviews and the initial seminars. The review of the literature presents the fact that not much prior research has been put into the specific area of the use of digital tools and technologies to facilitate the work and procedures of technicians performing maintenance on railway vehicles. Therefore, research and case studies from other industries have been used as a base for the frame of reference.

3.2.2 Interviews

Several semi-structured interviews with maintenance technicians and other personnel such as engineers and managers working in depots or with maintenance were performed as part of the qualitative research. This structure was chosen to understand the pain and desires of technicians and explore their thoughts by asking open questions and follow-up questions. The respondents willing to participate after the initial contact with the commissioners were contacted via email or phone to set up a time for the interview and depot visit. The respondents got information about the planned field visit and the interview in the form of an interview

information document. The interview information document consisted of the thesis description, purpose, and areas that would be discussed during the interview to prepare the respondents ahead of the visit and interviews, see Appendix A. The questions used during the interviews were formulated in an open way to gain fruitful insight, see the interview guide in Appendix B. The interviews were held in both Swedish and English and lasted for approximately 45 minutes and were held in person on-site. A total of 12 respondents were interviewed and a summary of relevant information regarding the interviews is found in Table 2.

Table 2: Interview details.

Respondent	**Role / Responsibility**	**Organization**	**Date**	**Duration**
A	Maintenance Team Leader	Workshop X	2024-03-05	00:55:00
B	Administrator	Workshop X	2024-03-05	00:20:00
C	Updating Maintenance Manuals	Workshop X	2024-03-06	00:45:00
D	Engineer	Workshop X	2024-03-06	01:00:00
E	Warranty Manager	Workshop X	2024-03-06	01:15:0
F	Maintenance Process Planer	Workshop Y	2024-04-11	01:00:00
G	Warranty Technician	Organization Z	2024-04-15	00:45:00
H	Maintenance Technician	Organization Z	2024-04-15	00:45:00
I	Mechanic	Organization Z	2024-04-15	00:45:00
J	Maintenance Technician	Organization Z	2024-04-15	00:45:00
K	Maintenance Technician	Organization Z	2024-04-15	00:30:00
L	HVAC Business Area Manager	Organization Z	2024-04-15	00:45:00

The interviews were documented via notes and an audio recording if approval was received from the respondent before the interview. The recording was used for transcription and analysis of the interviews and was later deleted after the thesis

project was finished. All respondents and data were anonymized to adhere to confidential handling.

Another interview in the form of a meeting was conducted with an expert in digital solutions at WSP. The meeting was set out to give deeper insights into possibilities and strains of utilizing digital tools and technologies, how one would develop and implement digital tools and technologies, types of data or information to implement, and how the technology is used today. The meeting was conducted 2024-05-02 on-site and lasted for 90 minutes.

3.3 Data Analysis

The data analysis of the interviews consisted of several steps. The audio recording of the interviews was first transcribed using KTH Play (Kaltura). The transcriptions were checked thoroughly and corrected if needed. The transcriptions were later coded using MAXQDA software in order to find relevant themes, codes and correlating areas among the different interviews performed. The results of the data analysis of the interviews are described in detail in Chapter 4 and the codes and themes established can be found in Appendix C.

4 Empirical Findings

This chapter presents the findings from the data collected during the interviews. To begin with, the maintenance procedures of a rolling stock depot are described. Thereafter, findings from the interviews are summarized and described in detail.

4.1 Maintenance Process of a Rolling Stock Depot

The description of the maintenance procedures is general and it is based on the combined findings from the different workshops visited and learned about. The organizations visited were workshops X and Y, and organization Z, consisting of consulting technicians working on different maintenance projects. The goal of the maintainer is to have enough vehicles maintained so that traffic can be run as intended. Fines can be issued to the maintainer if not enough vehicles can be put in traffic. Performing maintenance efficiently and with high quality is therefore of utmost importance. The procedures present in the maintenance process are divided into two primary divisions, Corrective Maintenance (CM) and Preventive Maintenance (PM), see Figure 5.

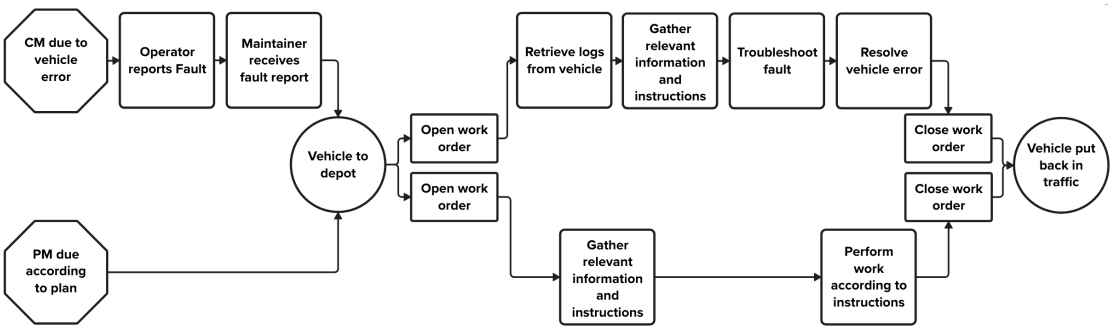


Figure 5: The vehicle maintenance process distinguished between corrective maintenance (CM) and preventive maintenance (PM).

The Corrective Maintenance process, CM, begins with a vehicle defect, failure, problem, or error of some sort. Errors or problems occurring during traffic will be reported by the driver operating the vehicle together with the TCC (Traffic Control Center). It could be an alert or error code appearing on the driver control screen of the vehicle, or that the driver senses that the vehicle is not operating as it should. Either way, the next step is to send the vehicle to the depot. After the maintainer has received the fault report and when the vehicle is in the depot, the fault report is put into an internal information system in the form of a work order. The work

order is later opened when the depot takes on that specific maintenance task. Thereafter, the first step is to take the logs from the vehicle to understand what has happened by analyzing the time of alerts concerning different parts and systems of the vehicle. The next step is to find and gather relevant information such as technical documentation regarding components and systems as well as schematics concerning electrical drawings and pneumatic drawings. These are found within another information system or portal in the form of different pdf documents. The next procedure is troubleshooting the fault by performing a visual inspection and testing different possible reasons for the fault occurring. For example, a multimeter could be used to check if a broken circuit can be the cause of the fault. The aim is to determine the root cause of the fault and solve it, and not just fix the vehicle failure, to ensure that the same fault does not reappear again. During corrective maintenance is the maintenance process documented internally to keep track of various interventions performed during the troubleshooting activities of a specific work order. This is to ensure that information and knowledge about a certain fault is transferable between different technicians working on that specific work order. When the maintenance is done will the work order be closed and information such as the type of maintenance performed, error codes if presented, interventions, personnel performing the maintenance, and time spent on the activity will be registered and saved. Lastly, the vehicle is put back into traffic.

The process of Preventive Maintenance, PM, begins when maintenance is planned and scheduled to be performed. Depending on the status of the vehicle, its system, and kilometers driven will the vehicle have specific time frames in which a certain maintenance activity needs to be performed and the vehicle will then be sent to the depot. The type of maintenance procedure to perform is planned and described in a work order. The next step is to open the work order and gather relevant information such as instructions or manuals, examples being information regarding how to change a specific part, disassemble a system and what tools to use. Checklists are used to confirm that the right work is performed correctly. When the maintenance is finished is the work order closed and the vehicle is put back into traffic.

4.2 Gather Logs from Vehicle to Perform Troubleshooting

Gathering the logs from the vehicle is one of the first actions taken in the depot while performing corrective maintenance. Today, as explained by several respondents, the possibility of receiving these logs remotely is not possible within their maintenance work as that functionality is not present on the vehicles. However, as explained by the Engineer, there are currently solutions for that on other projects where logs can be transferred prior to the vehicle arriving at the depot and be reached in real-time to assist with fault finding. It was mentioned by several respondents that logs are essential to have while troubleshooting a fault. The most important information needed from the logs were; the alert present when the problem occurred, the time stamp of the alert, a description of the experienced problem and additional information such as the speed of the vehicle or other alerts present. One of the maintenance technicians believes that getting logs in real-time could save time during maintenance as analysis of the logs could be performed without needing to go back and forth to the vehicle to retrieve logs.

4.3 Importance of the Right Competencies and Experience

Findings from the interview with the Team Leader and the Engineer indicate that there are currently maintenance staff shortages and staff changes. Further, another respondent describes how lack of maintenance staff can be an issue and another describes how it can also be a problem of not having the right competencies or experience of the whole staff, resulting in a stop in the maintenance process if someone is absent. Several respondents describe that great maintenance practices, and especially troubleshooting, stem from great experience, both in terms of time spent performing troubleshooting activities in general, but also time spent on a specific vehicle. If there are times when a new technician is present, the routine is often to pair them with a more experienced technician, to guide, support and teach the new technician. However, it is expressed that it takes a lot of time.

> *"... we go together through all the information, like this is how the vehicle is working, this is how the sub-components are working, this is how the components are working together. [...] so it takes time, I would say up to one year for a technician to understand how all the systems are working."*
>
> - Team Leader, Respondent A

4.4 Handling of Materials and Equipment

Finding material, such as spare parts, and equipment, such as a torque wrench, as well as having a lack of materials are areas that several respondents mentioned as frustrating while performing maintenance. Several respondents describe that a lack of spare components and materials in the depot can be the reason why vehicles are not in traffic. It was further explained how locating the right tools when needing them can be a challenge and that it is time-consuming. Respondents describe how the preparation of maintenance procedures is cumbersome. It was mentioned that finding what spare parts will be needed, if spare parts are in stock, and where they are located in the warehouse is time-consuming.

> *"... the process of finding the right material and where it is located has been cumbersome. If it is an electrical component do I need to go into the schematics first, find the component in the schematics, find the component number, copy that number to another list to see if it is in stock. [...] It is therefore many steps to know which component, where it is in the warehouse and if it is in stock."*
>
> - Warranty Technician, Respondent G

4.5 Types of Information, Documentation & Instructions used

It is clear from the interviews that several different types of information, documentation and instructions are used during maintenance work and procedures. Examples are technical documentation, such as vehicle descriptions, different schematics and diagrams, and maintenance manuals and instructions. This information and documentation is presented in a pdf format. The technical documentation of a system and its structure and function is usually described using text and 2D pictures. The maintenance instructions are also documented using text and 2D pictures such as a photo of a specific part or an assembly drawing using arrows and numbers indicating specific smaller parts of that system.

Both the Warranty Manager, the Team Leader and the Engineer described that apart from logs from the vehicle are electrical schematics, pneumatic schematics and technical descriptions of different systems and components mainly used when

troubleshooting faults. The Team Leader mentioned instances when the interpretation of schematics is lacking while accessing it as a pdf document on a laptop. This is due to the need to scroll through a big document with several pages. It was explained that there are instances when this documentation is printed out in an A3 or A4 format to visualize the content of the schematic and better understand it.

The information and instructions are often distributed digitally via an information system or portal, consisting of different hierarchy structures. Folders for, for example, instructions or technical descriptions are used and to find specific instructions is a tree structure used to filter according to sub-systems of the vehicle. However, it is explained that paper instructions are still used depending on the depot and specific vehicle or project. Several respondents explained that this can result in using out-of-date instructions and having a hard time locating and finding the right instruction, as a filtering process of a paper folder is needed. It was mentioned by respondents that having the instructions on a tablet, rather than on paper, would be preferred as it would be easier to access and quicker to go through.

4.6 Work Order Documentation

Several respondents described that documentation of interventions performed during the troubleshooting process is documented within an open work order. This is to assure that other technicians within the team can access current and relevant information about the fault they are trying to solve. However, the way the information is documented can be lacking and it can sometimes be hard to interpret the information given. It is described that the information written down is mostly specified in a "free text format" which, according to the Warranty Manager, can be troublesome as all information needed about the interventions and job performed can get lost in translation. However, according to one of the maintenance technicians, the free text format is a good way to document information as you can describe and explain however you please. Furthermore, you will not be restricted to document according to a specific format. Another maintenance technician emphasized the potential problem with having a too-restricted format in the form of mandatory fields to fill as some information can be unknown. The respondent responsible of updating maintenance manuals described that it is better to have too much information documented rather than having too little.

> *"... go deeper and explain, take your time to pass the information to the next person because they are going to definitely appreciate it. And maybe 15 minutes of your time is for them a couple of hours just trying to find the problem."*
>
> - Responsible of Updating Maintenance Manuals, Respondent C

It is also mentioned by the majority of respondents that the system or components identified as faulty as well as the interventions used to solve a specific fault are saved in the work order when it is later closed. This historical information is stored and can be gathered and used during a later troubleshooting process. It can serve as guidance when troubleshooting faults and the HVAC Business Area Mananger mentioned that this information can help them save time when troubleshooting, as much more information can be gathered about a specific error code or vehicle unit. The Engineer explained that they use specific keywords during the work order documentation and that these keywords can be used in the work order database to find historical faults of a specific component or system.

> *"So we can filter through that[historical word order data] and go through the historical faults and find similar faults and see what we have done there. [...] we have the access to come back to any single fault that we have performed on the train."*
>
> - Engineer, Respondent D

One respondent explains how access to this information depends on the vehicle manager and that it sometimes can be hard to reach and use. The respondent further explains that it is a weakness because using that information during troubleshooting is really helpful and relevant. Another respondent highlights how using historical work orders while troubleshooting can be beneficial and acknowledges that the searching and filtering functions should be better to find relevant historical work orders faster and easier. Filtering functions such as error code, vehicle, subsystem, or component are examples of useful filtering sections. However, the opinion of how often it is used and how beneficial it is differs. The Warranty Manager interviewed explains that it is not used very often, but usually when there is a more complex failure. Furthermore, it was expressed that it is mostly used by a more experienced technician or one with a more managerial role.

4.7 Finding Information within Different Information Systems

From interviews, it is clear that a lot of different information within different systems is used on a daily basis as a maintenance technician. Several respondents mentioned that the use of different information systems for different information can be tricky, confusing, and time-consuming to navigate. Furthermore, it is explained that it can be hard to find the exact information wanted. Respondents explain that in order to find the right instruction or manual, is a search tool used within the information systems or portals. However, the search used needs to be exact, otherwise will the information system not find it, making the process time-consuming and challenging.

> *"... the most time-consuming is to find all instructions and documentation, in my opinion. It takes a lot of time to sit and go through all systems and find the documentation needed."*
>
> - Maintenance Technician, Respondent J

The opinion of one respondent differs a bit, it was explained that you learn after a while how to navigate the information systems used to find the right information, instructions and manuals, but as a new technician, can the navigation of these information systems be hard and take a lot of time.

It was explained by one respondent that smarter and more interactive documentation has been used where clicking functions within the schematics would take you to the next page, making searches within these schematics easier. Another respondent mentioned that hyperlink functions between documents are being investigated to be incorporated. Furthermore, several respondents emphasize that links and connections between instructions and documentation can be helpful when trying to find the right information needed and to find it quicker.

4.8 Understanding Information and Instructions

Several respondents mentioned that the instructions used are not always clear, lack in quality and are not optimal. It was explained how the instructions can lack information in the form of what steps to use or clear descriptive pictures.

Furthermore, there are instances when the instructions are outdated.

> *"We have a lot of flaws concerning the maintenance instructions from the beginning really. There have been instances when notes have been put on instructions to adhere to a process which we think is better, because a lot of the instructions lack information. Then, the way we work is not according to the original document, we do it another way."*
>
> - Maintenance Process Planner, Respondent F

Furthermore, one of the interviewed maintenance technicians explains that it can be especially hard to follow instructions if they are not in their native language, which can be the case. Moreover, several respondents mentioned that reading and understanding electrical schematics is not easy for everyone and it is not a knowledge that everyone performing maintenance on vehicles possesses. It was explained that even for a more experienced technician, in terms of maintenance and vehicle knowledge, it can be troublesome to navigate the configuration of the vehicle using schematics. Furthermore, it was mentioned that the placement of components can be hard to understand, especially if you as a technician work with different vehicles daily.

Video instructions are mentioned by respondents as useful to better grasp and understand instructions. It is believed that video instructions can give more reassurance that you are performing specific tasks correctly and give more graphical context. Other respondents explain how video instructions can be useful in more complex cases when reading instructions is not enough to fully understand. Several respondents highlighted the fact that video instructions can serve as helpful for new technicians when trying to learn or prepare for a specific maintenance activity.

4.9 Interaction and Communication with Other Actors and Organizational Staff

When the maintenance technicians receive the fault report, the information presented can be hard to interpret. Respondents from all organizations agree that the information about a fault or an error reported by the driver is often lacking.

> *"... it [the information and description about the fault] varies. You have to respect that everyone is different [...] it can sometimes be hard to understand what they mean and therefore, where to start. It all depends on who has documented and reported the fault and how experienced they are."*
>
> - Warranty technician, Respondent G

Respondents explained that the fault report received could have gone through multiple actors before reaching the maintainer. It was explained that information can get lost due to this. The Warranty Manager, explains that if the fault reporting is of high quality and all relevant information is present such as error code, vehicle behavior, and time stamps, time can be saved during the maintenance process. The troubleshooting can start directly and technicians can skip the step of retrieving the logs from the vehicle if thorough fault reporting is received.

It was expressed that communication between the maintainer and stakeholders such as the vehicle manager, the vehicle manufacturer and sub-suppliers are activities performed in the maintenance process. Several respondents described that they would get in contact with sub-suppliers about software faults in the systems provided by them. Especially since there is a lack of software knowledge of technicians, but also since the maintainer is not always allowed to touch or alter specific components or systems, according to contractual agreements.

> *"... they [component and part suppliers] are experts in making software, creating the code for the software. [...] for some things, of course, we can see the code, but first, we are not allowed to touch the code, we don't have the software to change the code, and probably we don't have the knowledge to know what to look for in the code."*
>
> - Team Leader, Respondent A

As expressed by the Warranty Manager, one of the first steps, if a system failure is present in a component, is to exchange the component if possible, but that requires the availability of spare components. The faulty component is then sent to the sub-suppliers to let them investigate the root cause of the fault present.

The Engineer expresses that taking out a component and sending it to the sub-supplier is not always optimal. It can take several months to ship the component to the sub-suppliers and get it fixed, and if no spare components are available, the vehicle will be at a standstill in the depot. Therefore, depending on the fault or the situation would they work together with the sub-supplier to try and find the solution to that fault or get recommendations on how to solve a certain problem directly in the depot. The engineer expressed that they would email or phone sub-suppliers about sub-systems and components to understand the functionality better or get help with analyzing the logs and how to fix the components. The Team Leader further explained that all documentation and information about the component or system can be unavailable depending on the supplier, making the communication with sub-suppliers needed to receive more in-depth information. Furthermore, there are instances when the vehicle manufacturer helps and guides the actor performing preventive maintenance even if the warranty period of some vehicles has passed. The communication can however be tricky as things can get misunderstood.

> *"... maintainer comes to us and say "we are having this fault on the unit [...] we don't know what is going on, we did this and this step, but we can't find a fault." We help them [...] sometimes by phone, but with virtual interaction, I think everything is easier for not misunderstanding things."*
>
> - Engineer, Respondent D

Instances when other actors are contacted for help, such as other maintenance depots or organizational colleagues, also occur as the right experience, competencies and information are not present in the depot. Communication and collaboration with people with more experience via mail or telephone are then needed to move forward in the maintenance process, but giving thorough guidance can be hard as much information and context can get lost.

4.10 View on Helpful Tools and Technologies in the Future

When asked about what would be helpful to have while performing maintenance on railway vehicles it was mentioned by one respondent that a big information portal

with all relevant information gathered would be beneficial. Furthermore, functions to give guidance regardless of the fault or error code present would be of tremendous help. Moreover, it was explained by several respondents that the ability to get guidance on where to start and possible root causes while troubleshooting will serve as very beneficial.

Several respondents imagine the potential of using VR in the future of maintenance. One respondent sees potential in using it while training personnel without slowing down or needing to stop the production or maintenance process. Another respondent sees potential in using it to familiarize oneself with the vehicle remotely, before getting to the depot, ultimately preparing with knowledge of the vehicle configuration. Lastly, another respondent believes it is a better way to learn and interpret instructions compared to reading.

> *"... I believe in general that technicians have trouble reading and then understanding, it [learning and comprehending instructions] is much more visual and the ability to touch and feel."*
>
> - Maintenance technician, Respondent K

Some respondents also mentioned AR as a potential application in maintenance practices. One respondent believes that AR can be helpful for educational purposes. Other respondents see potential in using it while performing maintenance procedures as valuable information can be displayed in front of the technician's view. Furthermore, the fact that AR applications and AR goggles can be used to help with remote guidance of maintenance by connecting experts to the technician was highlighted by respondents. One respondent further explained how that form of communication can serve as beneficial as the language barrier will be smaller since the interaction and communication will be more visual.

The overall view of the respondents to implement digital tools and technologies to their work and procedures was positive, however, one respondent was more hesitant to new implementation and emphasized that the routines used today work and that you just need time to learn how to go about it.

5 Analysis and Discussion

In this chapter, the two sub-questions are answered by analyzing and discussing findings from the empirical findings and literature. The sub-question: What are the main challenging activities for technicians performing maintenance on railway vehicles? is answered using the findings from Chapter 4. The sub-question: How can various digital tools and technologies be used to facilitate technicians performing maintenance on railway vehicles? is answered by analyzing the different digital tools and technologies presented in the literature, Chapter 2, and discussing their ability to facilitate the challenges identified.

5.1 Main Challenges and Time-Consuming Activities

The findings from the interviews resulted in the identification of several challenges and time-consuming activities. These challenges and time-consuming activities are displayed in Table 3 and are individually explained and analyzed in detail below.

Table 3: Main challenges and time-consuming activities.

Challenge area	Challenges and time-consuming activities	Challenging activity	Time-consuming activity
Find and Access Information	Navigating different information systems and finding all relevant information	•	•
	Locating material and equipment	•	•
	Needing vehicle in depot to retrieve logs		•
Understand Information	Understanding instructions	•	•
	Understanding placement of parts	•	•
Support, Communication and Experience	Know where to start when having a fault	•	•
	Relying on information and help from experts	•	•

From the challenges and time-consuming activities displayed in Table 3, was the challenge *Navigating different information systems and finding all relevant information* the most emphasized one, and challenge *Know where to start when having*

a fault was the second most mentioned by respondents. *Needing the vehicle in the depot to retrieve logs* was considered to be more of a time-consuming activity rather than a challenge, but simultaneously something mentioned by several respondents.

The challenges and time-consuming activities are present in several parts of the maintenance process. A mapping of the maintenance procedures together with the challenges and time-consuming activities described in Table 3 is presented in Figure 6 to better understand when in the process these challenges and time-consuming activities arise.

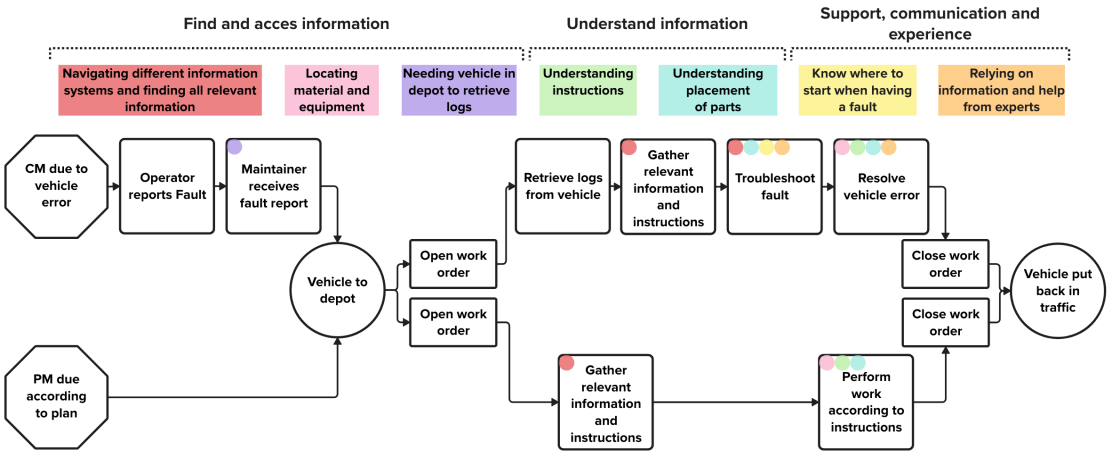


Figure 6: Mapping of the maintenance process and the challenges and time-consuming activities.

5.1.1 Navigating different information systems and finding all relevant information

As presented in the empirical findings, several different information systems or information portals are used while gathering the information needed, such as technical description, maintenance instructions, and inventory status of components, in order to perform maintenance procedures. This way of structuring information has its downside as technicians need to be able to navigate and access several different information systems or portals simultaneously. Checking if spare parts are in stock to be able to perform a specific maintenance activity is handled in another system than the system with the instructions, making the preparation of maintenance procedures time-consuming. Furthermore, it was explained in the interviews that it can be hard to find the specific instruction needed as you need to know in which folder to look for it. It is expressed that having search functions

for finding the right information is helpful, but it is sometimes hard to find the right instructions as you need to be very precise with the search words used. This process is even more challenging while using paper documentation and not accessing the information digitally. All these factors make the process of finding the right information rather hard and time-consuming, especially for less experienced technicians.

5.1.2 Locating material and equipment

Locating material and equipment needed to perform maintenance is also considered a challenge and time-consuming activity as equipment and tools are being misplaced. Furthermore, navigating the warehouse to find spare parts and components is also expressed as challenging, especially if you have less experience in that depot.

5.1.3 Needing vehicle in depot to retrieve logs

Another time-consuming activity is the time waiting for the vehicle to arrive at the depot to reach the logs and operational data. The analysis can only begin once the vehicle is in the depot and logs can be retrieved. If diagnosis is started earlier, spare parts, equipment and instructions can be prepared and the actual maintenance work can begin as the vehicle reaches the depot, resulting in a more efficient maintenance process.

5.1.4 Understanding instructions

It was mentioned that there are instances when instructions can be challenging to understand. This is due to the lack of quality of descriptions, instances of outdated instructions and the structure of presenting the information. It was expressed that it is not always clear at the beginning of instructions what material or equipment will be used. Furthermore, understanding the instructions while reading them is expressed as challenging and it also takes time.

5.1.5 Understanding placement of parts

The reality of having different versions of a vehicle and working with different vehicle types is troublesome in terms of being well experienced with the configuration of the vehicle, ultimately finding the right component or system in a fast

and easy manner. The ability to navigate electrical schematics and understand the connections, configuration and the actual placement of parts and components on the vehicle is expressed as important while troubleshooting. It was expressed that it could be especially challenging for less experienced technicians to interpret, and the more experienced technicians said that reading electrical schematics was often fairly easy, but it can take time.

5.1.6 Know where to start when having a fault

The troubleshooting process is considered challenging. It was expressed by respondents that troubleshooting and skills related to corrective maintenance stem from much experience. You are more likely to be able to know where to start when having a fault if you have worked with similar vehicles or similar faults in the past. Furthermore, it can be tricky, even for more experienced technicians, to investigate and solve a fault, especially since there are not often clear instructions on how to solve a fault. Moreover, if the fault report lacks information, it can be challenging for the technicians to start the troubleshooting process. Much is unknown and you need to go by trial and error, which can be both challenging and time-consuming.

5.1.7 Relying on information and help from experts

There are instances where all information about a system or components is not available because troubleshooting of that component is handled by others. Then technicians rely on suppliers to perform the proper troubleshooting of a component or system. This is considered to be both challenging and time-consuming as the maintenance process for the technician can come to a halt and they either need to send the component to the supplier or have the supplier come to the depot, ultimately a process that hinders the vehicle to be put back into traffic. It was described that communication with the suppliers can take a long time since it consists of a lot of emailing back and forth. Moreover, it can be hard to communicate in general as information can get lost and language barriers can hinder the process. Furthermore, not having the right experience or competencies needed in the depot to perform different maintenance procedures is also a challenge. Communication and collaboration with stakeholders and experts is therefore the case while performing maintenance, but it is not always considered easy and time efficient.

5.2 Area of use of Digital Tools and Technologies

To distinguish what digital tool or technology could be beneficial to implement and how it could be used to facilitate the work and procedures of maintenance technicians are benefits, drawbacks and effort mapped for the different technologies presented in the Frame of Reference, Chapter 2. The effort needed to develop and/or implement is based on time and resources needed as well as if the technology is used on the market today. The analysis of the digital tools and technologies is based on a combination of literature and findings from the Digital Solutions Expert interview, and it is summarized in tables for each technology area.

5.2.1 AR Technology

Table 4 presents a summary of potential usage areas of AR technology and is further discussed below.

Table 4: Mapping of AR Technology.

AR	Benefits	Drawbacks	Effort
AR for remote support	- Access to expertise remotely - Access via tablet or HMD	- Relying on expert to be available - HMD can be heavy and unpractical	- Medium to high effort to develop
AR for instructions	- Augmented instructions - Focus on maintenance activity - Lowering knowledge level needed	-HMD can be heavy and unpractical	- High effort to develop
AR for navigating vehicle	- See behind panels during maintenance procedures - Lowering knowledge level needed	- HMD can be heavy and unpractical	- High effort to develop

AR for remote support:
According to several authors [4], [10], [13] remote support is one advantage of using AR in maintenance. By incorporating an AR system with a device for the technician, experts, regardless of their location, can see what the technician is seeing and give real-time remote guidance via voice and augmentation in the

technician's device. The device used by the technician can be either handheld, like a tablet, or head-mounted, like AR goggles, the crucial part is that the device can record in a high resolution so that the remote expert can see the technician's environment. Furthermore, tracking will also be a part of this solution as guidance given by the expert in the form of, for example arrows, will be tracked to the object and not be placed statically in the device used by the technician. The effort to develop this system is considered to be medium or high depending on how advanced the solution is [28]. If augmented information can be given in text, arrows or animations, or if the technician can interact with the augmented information as well [25]. Furthermore, it is also dependent on how automated the system is. A lower development and implementation effort will be needed if less automation is incorporated, and if the system and the information presented is controlled by the expert. One drawback considered is that this technology implementation is reliant on the availability of experts when support is needed. Furthermore, HMDs used, such as AR goggles, can be experienced as heavy and impractical [29].

AR for instructions:
Another area where AR technology can be used in maintenance is while performing tasks or procedures according to maintenance manuals by receiving additional context information [26] augmented onto the real environment in real-time by using different devices [10], [13], [25]. Case studies of using AR in the aviation industry [35], [37] to perform procedural maintenance tasks are present. The results from the case studies show that the process became more efficient, both in terms of time, but also the quality of the process and its results [35], [37]. However, this type of technology can be harder to implement when it comes to corrective maintenance when the exact cause of the fault is unknown and no clear instructions can be used. The Digital Solutions Expert mentions that one could go about and build this AR application using a game engine, however, to have the AR technology working as intended, there is a need for having an advanced tracking system of the HMD and storyline features to track what the technician is doing and display the information needed for each step, ultimately making the level of effort to develop and implement high. Furthermore, according to authors [24], there is still development to be done to achieve great results in terms of the reliability and usability of AR applications.

AR for navigating vehicle:
Another useful area of AR could be the ability to navigate the vehicle configuration and look behind panels during maintenance procedures to easier find specific components. While using an AR solution together with a HMD, such as goggles, technicians can work on the actual vehicle simultaneously as they receive augmented information about the placement of parts. As one case study shows, the use of a 3D information model can be used in an AR solution to highlight components hidden behind panels and walls [38]. However, this solution needs advanced tracking to be able to take in the placement of the vehicle and combine that information with the 3D model to augment the right information at the right placement.

5.2.2 VR Technology

Table 5 presents a summary of potential usage areas of VR technology and is further discussed below.

Table 5: Mapping of VR Technology.

VR	Benefits	Drawbacks	Effort
VR for training and education	- Immersive training - No interuption of maintenance process	-Motion sickness - HMD and controllers needed	- High effort to develop
VR for visualization	- Immersive visualization	- Motion sickness - HMD and controllers needed	- Medium effort to develop

VR for training and education:
According to the literature, there are some areas where VR technology is more applicable and suitable in the field of maintenance. Some examples are education and training of personnel [10], [30]. Furthermore, according to the Digital Solutions Expert, the application area of VR is to give an immersive experience where you can interact digitally with an object. The idea is to be able to fully interact with a digital version of the asset and interact with different parts and components. Game engines can be used to develop such an application. By incorporating CAD models of the vehicle and its parts together with interactive functions with the vehicle, an immersive world can be built. However, in order to use VR and fully

interact with the vehicle according to specific steps in an instruction or maintenance manual, an extensive development process is needed. The Digital Solutions Expert mentioned that it is one thing to build a virtual world where one can explore the visualization of a vehicle but to be able to interact freely with parts and components, a more advanced development process is needed, such as storyline development. Furthermore, one downside to using a more immersive technology is motion sickness. Authors mentioned that motion sickness was experienced while using VR technology [8], [24].

VR for visualization:
Using VR for visualizing and navigating the vehicle will not require storyline development, as one would not need to follow specific tasks and have the system determine if they are performed correctly. It can be a good way of exploring the vehicle while not having access to a vehicle in the depot or taking time and effort to move heavy objects and parts of the actual vehicle. However, an HMD with controllers is needed in order to interact immersive with the vehicle, which can result in motion sickness [8], [24].

5.2.3 AI Technology

Table 6 presents a summary of potential usage areas of AI technology and is further discussed below.

Table 6: Mapping of AI Technology.

AI	Benefits	Drawbacks	Effort
AI for analyzing, searching and filtering	- Finding information faster and with help	- Establishing data structure - Training of AI model required	- Medium to high effort to develop

AI for searching and filtering:
Authors [18], [20], and the Digital Solutions Expert highlight the benefits of using AI while searching for information. The Digital Solutions Expert mentioned that there is potential in implementing AI technologies to better search, find and filter information. By enabling an AI add-on to the information systems used, could technicians get help in finding information quicker [20]. With a smart AI

implementation, the challenge of using the right search words will be minimized. However, data structure needs to be established as well as having access to data from different information systems, and sufficient training of an AI model [20]. Therefore, it is considered that the effort to implement depends on the availability of the data needed and also the level of how advanced the system is. The advancement of information systems used today differs, but adding a more advanced searching and filtering function, better than the ones used today, could be possible. Furthermore, this technology does not require additional hardware implementation.

5.2.4 Smart Documents Technology

Table 7 presents a summary of potential usage areas of Smart Documents technology and is further discussed below.

Table 7: Mapping of Smart Documentation Technology.

Smart Documents	Benefits	Drawbacks	Effort
Interactive schematics	- Connections to information - Highlighting effects	- Not the standard PDF format used today	- High effort to develop and implement

Interactive schematics:
As the context and information within the schematics can be rather hard to grasp while accessing them via a pdf file, especially if it is in paper format, there is a potential benefit of visualizing the same schematics in a digital interactive format. This way of visualizing the information can enable a better overview as well as a better interpretation of the connections of components and parts studied in the schematics. As described by one case study [31], contextual information between diagrams, such as highlighting and spatial information, can facilitate the troubleshooting process of technicians as it enables them to perform their tasks in a more efficient manner. However, to implement this technology will the development of new interactive documentation be needed to achieve useful functions such as highlighting of components and links to contextual information.

5.2.5 RFID Technology

Table 8 presents a summary of potential usage areas of RFID technology and is further discussed below.

Table 8: Mapping of RFID Technology.

RFID	Benefits	Drawbacks	Effort
RFID tags	- Tracking of objects - Well used technology	- Tagging objects manually	- Low effort to implement

RFID tags:
RFID technology can be used to track different objects [2], [19] and there is therefore potential to use this technology to keep track of tools and equipment needed to perform maintenance. This technology is developed and used in industry [19] and is therefore regarded as easy to implement. Requirements will be to invest in the right system and hardware. The drawback is that objects need to be tagged manually in order to track them.

5.2.6 Digital Twin, Digital Shadow and Digital Model

Table 9 presents a summary of potential usage areas of Digital Twin-, Digital Shadow- and Digital Model Technology and is further discussed below.

Table 9: Mapping of Digital Twin, Digital Shadow and Digital Model.

Digital Twin, Digital Shadow & Digital Model	Benefits	Drawbacks	Effort
Digital Model	- Digital visualization	- Model not representing real status	- Medium effort to develop
Digital Shadow	- Digital visualization - Real-time data transfer - Ability to work proactive	-Improved ICT needed	- High effort to develop and implement
Digital Twin	- Digital visualization - Two-way real-time data transfer - Ability to work proactive	- Advanced analytics system and improved ICT needed	- High effort to develop and implement

Digital model:
As previously mentioned, the different integration levels of digital representations, distinguished by the data flow between the digital and the physical asset, influence the level of effort needed to develop and implement. The first level, the digital model, can enable a better contextual understanding of the asset as more visual graphics are incorporated. As the use of engineering knowledge, such as information about the design of the asset and the physicality of it, is explained to not be utilized during the operational phase of an asset [10] potential is seen to use this information to give more contextual understanding of the asset. Furthermore, the ability to easier navigate the asset and investigate different layers of physical information. For example, cut views or rotational functionalities of a CAD model can be used to better grasp the configuration of the vehicle without having the vehicle in the depot or actually taking the time to remove components. Furthermore, as mentioned by [8] other relevant data and information such as technical documentation can be linked to the 3D model. This implementation is considered to be not too complicated since CAD models of the vehicle from the development phase exist in the hands of the vehicle manufacturer. However, there is a question of whether this documentation can be shared with the vehicle manager and the vehicle maintainer without overstepping any informational privacy. Furthermore, to have information linked to the digital asset, such as technical description, there is a need for a system that can handle that interaction and it was mentioned by the Digital Solutions Expert that access to a centralized data platform would be needed.

Digital Shadow:
The next level of integration, the digital shadow, can serve as beneficial technology to implement as data from the physical asset will be automatically transferred to the digital asset [8]. Information about the current status of the vehicle can be received while it is in traffic and analysis of vehicle behavior can be performed. However, to utilize this technology, the vehicle needs to have developed ICT systems to handle this data transfer. It puts requirements on the communication technology existing on vehicles today. As mentioned by Scheffer et al. [13] is the data transfer systems today not up-to-date. This makes the implementation and development of this technology rather complex and more parties are needed to be involved such as the vehicle manufacturer, the vehicle manager as well as responsible parties for wayside communication.

Digital Twin:
By having a digital twin can real-time information from the vehicle be accessed to better plan and predict maintenance by using a digital asset to analyze and simulate vehicle behavior [8], [10]. Furthermore, it is a way of controlling the actions of the physical asset by using the digital asset [8]. This can be especially useful when access to the physical asset is limited [23]. Furthermore, the use of this technology can nurture a transfer to a more preventive maintenance approach, which is viewed as beneficial both in terms of efficiency and resources [2]. However, supporting technologies are needed such as sensors on the vehicle, IoT, data analytics and advanced ICT to create two-way automatic data transfer, making the effort to develop high [21].

5.2.7 eMaintenance Technology

Table 10 presents a summary of potential usage areas of eMaintenance technology and is further discussed below.

Table 10: Mapping of eMaintenance Technology.

eMaintenance	Benefits	Drawbacks	Effort
eMaintenance system	- Real-time data transfer from asset - Technology used today	- Vehicles with developed ICT and sensors	- Medium effort to implement

The literature presents that eMaintenance technologies are used today in the maintenance of railway vehicles [2], [10] to make informed decisions in regards to planning and scheduling of maintenance, but can also be used during troubleshooting procedures [17]. By getting information from sensors and systems to an information portal, one can monitor and diagnose vehicle status and behavior while it is in traffic. The effort to implement this technology is considered to be medium as it is a technology used today, but the implementation of ICT systems and sensors will be needed on vehicles to achieve real-time data transfer.

5.3 How can Digital Tools and Technologies Facilitate the Challenges Identified

From the initial analysis of the main challenges and time-consuming activities, and the area of use of digital tools and technologies, potential digital technologies and tools are mapped to the challenges identified, see Figure 7. As seen in Figure 7, some challenges and time-consuming activities have the potential to be facilitated by a number of different tools and technologies, for example, *understanding placement of parts*. However, potential technology implementation is delimited to one or two for other challenges and time-consuming activities.

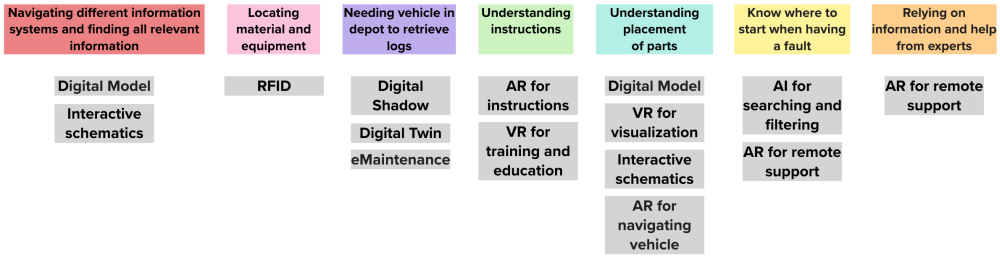


Figure 7: Potential digital tools and technologies to facilitate challenges identified.

To better analyze the potential trade-off, of the digital tools and technologies presented, an evaluation matrix was developed consisting of an estimation of the benefit given and the effort needed to implement the different digital tools and technologies discussed in section 5.2, see Figure 8.

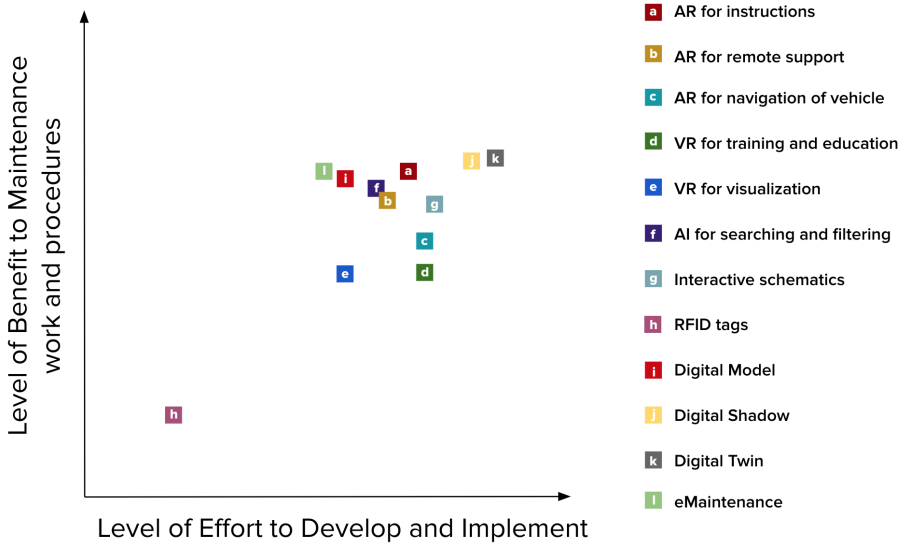


Figure 8: Evaluation Matrix of potential trade-off between benefit and effort of tools and technology concepts.

The effort consists of an estimation combining factors such as time and resources needed to develop and implement, and if the technology is used on the market today. What digital tools and technologies could be most beneficial and how they could be used to facilitate the challenges and time-consuming activities identified are analyzed and discussed below.

5.3.1 Navigating different information systems and finding all relevant information

As finding information is regarded as challenging, there is a potential to use technologies that gather relevant information. By using a 3D information model of the vehicle including relevant data linked to its systems, components and parts can the technicians find information by identifying a system in a 3D model rather than knowing the exact name to search for, or in which folder to check. While standing in front of the vehicle at the placement of a faulty component, technicians can interact with a Digital Model and click on a component to gather relevant information such as instructions, technical descriptions or what spare parts to use and if they are in stock. The development and implementation are considered to be medium since existing CAD files can be used. The potential trade-off is therefore considered to be high, see Figure 9. It could be argued that a Digital Shadow or Digital Twin could aid as helpful when needing to find the right information and documentation as it also can serve as a centralized information base. However, the integration level and the flow of data between the digital and the physical asset that these technologies realize, compared to the Digital Model, may not facilitate in finding the right information. It is therefore considered that implementing a Digital Twin or a Digital Shadow for this particular challenge will not give a great trade-off since the effort needed will be higher than the benefits realized, hence they are not considered a viable solution.

It was expressed by respondents that linking functions within schematics were used to track connections between pages of the schematics, which was considered helpful when trying to find the right information. It is therefore reasonable to believe that interactive schematics with other linking functions can serve as beneficial for the technicians when finding relevant information. The search for information can be less time-consuming if smart linking functions, with links to other relevant information, can be incorporated into the schematics. With the possibility to link

information such as inventory, technical documentation, and instructions related to a component in the schematics, technicians can more easily gather relevant information without going through as much documentation, and less time and frustration will be put into navigating different information portals. However, the development and implementation needed are considered high as a new schematics format with interactive schematics and a visualization system will be needed. A lower effort is considered needed if linking functions can be incorporated into already existing pdf schematics. However, the ability to implement such linking functions is still a question. Literature briefly suggests that there are possibilities to link information to a 3D model or digital model [8] or to components of the interactive schematics if it is visualized and developed in a specific system [32], but no clear cases in which this can be achieved in an already existing pdf document have been found. Furthermore, using this technology can make technicians working from schematics find information quicker, but it is not as useful for maintenance procedures when schematics are not used. It is therefore considered that the possible solution of incorporating new interactive schematics includes much development and implementation with benefits for only corrective maintenance where schematics are more used, making the potential trade-off low, as can be seen in Figure 9. More research is needed to find tools and technologies that can be used to add linking functions to already existing pdf schematics so that a lower effort to reach the benefits of linking functions can be achieved.



Figure 9: Evaluation matrix of digital tools and technology concepts relevant for the challenge of: Navigating different information systems and finding all relevant information.

Findings from the interviews show that the access to the relevant information and documentation needed varies, as some respondents expressed that paper is still used and others stated that information and documentation are accessed digitally via a computer. With this in mind, one rather simple digital tool to implement, that can facilitate the maintenance procedures of technicians, is a tablet. With a tablet, technicians can have access to the information needed regardless of the position in the depot. Furthermore, the filtering and navigation of documentation will be easier as it is accessed online where searching functions and linking functions between pdf pages of schematics can be used to find information more easily. Moreover, it will be easier to access updated instructions compared to when information and documentation in paper format are used. This implementation is not new, or technology intense, but as interview findings suggest, it can be an investment where the trade-off is considered big. The technicians do not need to take time to walk to folders or stationary computers to access the information and they can more easily filter and search for the information needed.

5.3.2 Locating material and equipment

The challenge of keeping track of material and equipment can be helped by using supporting technologies such as RFID. By placing tags on equipment and using tracking devices, the location of the equipment can be found regardless of placement. Furthermore, the same tags can be used in the warehouse to easier find which shelf to go to in order to find the right spare parts or components needed. However, one could argue that this challenge would not be present if tools and equipment were put back in their rightful place and if the warehouse were structured and neat. If equipment and tools are stationed at convenient places and put back when not used, the challenge and time spent searching for them will be diminished. Even if the technology is considered to be rather easy to implement since it is used on the market today, the question still arises of whether it would be worth the investment, as time will still be spent on locating the equipment, even if the time spent can be shortened, see Figure 10. It is therefore considered that using solutions within the area of lean practices and 5S methods can be a way of solving the challenge of locating material and equipment as tools will have their place and will not be misplaced and the warehouse will be structured and clean [5], [6]. It is therefore considered that by implementing these practices can

successful results be met with a lower investment in new technology.



Figure 10: Evaluation matrix of digital tools and technology concepts relevant for the challenge of: Locating material and equipment.

5.3.3 Needing vehicle in depot to retrieve logs

With a Digital Shadow or Digital Twin, the time-consuming activity of waiting for the vehicle to arrive at the depot to access the logs can be diminished. The time waiting for the vehicle can be put towards analyzing the logs and establishing the root cause of the fault. Furthermore, the time can be used to better plan the troubleshooting activities and prepare relevant documents, spare parts and equipment. However, a Digital Twin or a Digital Shadow are rather advanced technologies where the combination of engineering technologies such as the design as well as the operation of the asset is used to analyze data and make informed decisions. Another less advanced technology is eMaintenance solutions where real-time data transfer is realized [10], [2]. Data is then transferred in real-time from the physical asset to an information system, rather than a digital asset [10], [2], [17]. Furthermore, as this technology is used in projects today, the effort to implement it in other projects can be considered not as drastic as the one needed for a Digital Twin or a Digital Shadow [17]. Furthermore, one advantage of a Digital Twin is the ability to control the physical asset with the digital asset, if the physical asset can be hard to reach. In this case, it is considered unnecessary as the asset is moveable and can be reached compared to other assets placed far away. It is therefore considered that the challenge of needing the vehicle in the depot to retrieve logs can be solved using real-time data transfer from the asset to an information system,

rather than a digital asset, with less investment in new technology compared to Digital Twin or Digital Shadow, see Figure 11. However, for this implementation to fully facilitate the maintenance procedures of technicians, it is fruitful to use the time saved to carefully analyze the logs and solve the root cause of the failure, to ensure that the efficiency and the quality of corrective maintenance practices are enhanced.



Figure 11: Evaluation matrix of digital tools and technology concepts relevant for the time-consuming activity of: Needing vehicle in depot to retrieve logs.

Furthermore, other potential benefits can be seen while utilizing this real-time data transfer in the planning and scheduling of preventive maintenance, transforming the maintenance strategy towards a more predictive one, as discussed and empathized in [17]. The way of optimizing the planning of preventive maintenance could make the maintenance process more efficient in terms of having the right resources, but also performing maintenance before faults occur.

5.3.4 Understanding instructions

The use of AR solutions with an HMD, such as AR goggles, while performing work according to instructions, can be helpful as it can provide technicians with augmented information such as animations, static 2D/3D models and text labels, at the right time making the reading and understanding instructions less challenging. Moreover, as instructions can be given in an immediate and immersive manner during the maintenance procedure the level of knowledge and competencies needed to perform such tasks can be lowered, which is positive when handling

shortages of staff and the lack of knowledge and competencies. However, in order to have a fully functioning AR system, that gives specific augmented instructions in real-time, is a development process needed in the form of creating a storyline so that the system can recognize tasks as performed and give information about the next task. Furthermore, it is also a question of acceptance among the technicians to use this new and rather drastic technology implementation. The overall view of implementing new digital tools and technologies was positive according to the interviews. However, it was stated that the solution needs to be easy to use and depending on the HMD used could motion sickness and strain on the ergonomy be a factor. Furthermore, according to the Digital Solutions Expert, a lot of time and money would be required to develop such a system. However, according to authors [37], [35] higher results are achieved while using AR for instructions in terms of faster time spent on the procedure and better quality of the procedure performed. It is reasonable to think that the same results can be achieved while incorporating AR in the maintenance procedures of railway vehicles. It is therefore considered that the effort needed is high, but the benefits achieved are also high as time spent on maintenance activities and the struggle to understand and interpret the instructions can be minimized, see Figure 12. Simultaneously, it is a great way of handling the lack of personnel with the right knowledge and competencies as the knowledge level needed to perform the maintenance correctly would be lowered. Less time needs to be spent on training and educating technicians before maintenance practices and more experienced technicians do not need to spend that much time educating and guiding new technicians during maintenance.



Figure 12: Evaluation matrix of digital tools and technology concepts relevant for the challenge of: Understanding instructions.

Another area of technology to use to tackle the challenge of understanding instructions is VR. Respondents see potential in using VR to train less experienced technicians in a safer environment ensuring that no actual vehicle is tampered with. This can enable faster and more accessible learning and understanding of instructions for less experienced technicians as it can be done regardless if a vehicle is present in the depot or not. Furthermore, it is also a more immersive learning experience compared to reading or studying maintenance manuals prior to the work in the depot. However, according to the Digital Solution Expert, more resources are required to develop and build such a solution than traditional manuals. Furthermore, this use of technology could help with training of technicians making the understanding of instructions and procedures better, but the actual use of it while on the workshop floor performing maintenance is not that beneficial. It is therefore considered that the trade-off between investing in this technology and the actual use to help with the understanding of instructions while performing maintenance is rather low, see Figure 12. It is therefore reasonable to believe that more value can be achieved in using VR technology to educate technicians before the actual maintenance work and procedures. In the long run can the implementation of VR help technicians understand the instructions better, but to realize positive results on the efficiency of maintenance procedures in a quicker manner is AR for instructions recommended.

5.3.5 Understanding placement of parts

One way of solving the challenge of not understanding the configuration of the vehicle and the placement of parts is by having access to a 3D model of the vehicle. This can be developed using different technologies and development software, examples being game engines or OpenGL/WebGL. By having access to a 3D model of the vehicle, technicians can explore the vehicle and its parts to help find components, understand the connection and relation between components, and look behind panels. It is reasonable to believe that this can help technicians understand the placement of parts and time can be saved to locate parts before working on the vehicle, ensuring that no extra work is put into the removal of wrong parts. The question is however if this information should be accessed virtually using VR technology or if a digital model accessed via a tablet or a computer is enough. On one hand, VR can enable a more immersive experience and one would explore the

vehicle more as you would in reality, but on the other hand it will require a lot of development and also more hardware. Furthermore, it will not be as quick and easily accessible compared to accessing the same information using a digital model via a computer or a tablet as VR goggles will need to be used in a safe area. Moreover, it is considered less user-friendly as motion sickness is often experienced [24], [8]. Therefore, a digital model is considered to have a better trade-off, see Figure 13. One could also argue that with the use of AR, as in the case example by Xie et al. [38], can the understanding of the placement of parts be helped by having the ability to see through panels while standing by the vehicle using AR goggles. This solution is, therefore, more immersive than accessing a Digital Model on a tablet, and can be used at the vehicle while performing maintenance compared to using VR. However, the case study presenting the use of this technology included a building as an asset instead of a vehicle, hence the building is statically placed and the vehicle is movable, putting pressure on the tracking system of the AR solution. Furthermore, the effort for this implementation is considered high as a 3D model of the whole vehicle is needed together with an AR system, see Figure 13.

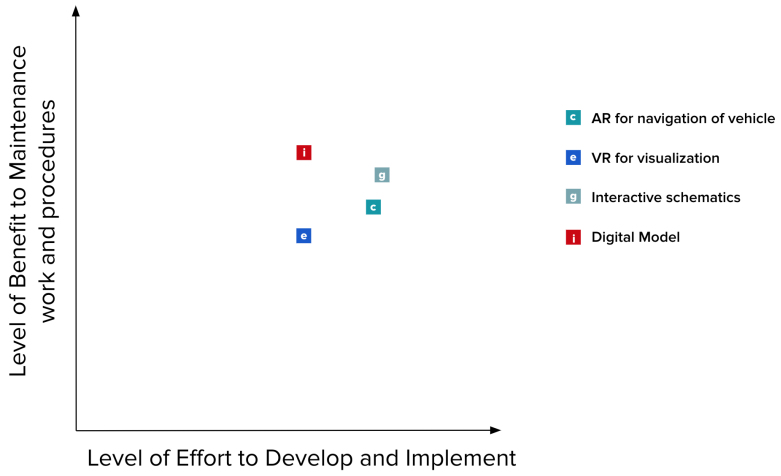


Figure 13: Evaluation matrix of digital tools and technology concepts relevant for the challenge of: Understanding placement of parts.

By using digital technologies to create interactive schematics, for example the tool SDViz [31], the challenge of understanding the placement of parts can be minimized and the level of knowledge and competencies to interpret schematics can be lowered. By having highlighting functions, the navigation of schematics can be more efficient as the technician's attention will be drawn directly to the position

needed and relevant connections within the diagram. Furthermore, and explained by technicians are linking functions, connecting wires between pdf pages of the schematics, helpful when reading and understanding them. However, interactive schematics or the linking functions explained to be used within schematics, are solutions that may help with navigating the schematics faster, but translating the information within the schematics to the actual vehicle can still be troublesome as no visual context is given. Yang et al. [32] point to the fact that understanding the placements of parts can be helped by using interactive schematics paired with links to 3D models to have both a holistic understanding of the connections, but also the actual layout of components in reality. However, the development of new schematics will be required as the ones used today are in a pdf format. Compared to only using a Digital Model requires this solution more development as new interactive schematics, highlighting functions, linking functions and connections to a 3D model will be needed. It is therefore recommended to incorporate a Digital Model to visualize and explore the configuration and connections of the vehicle as it will help the technicians understand the placement of parts and give additional contextual information to the already existing schematics but will require a lower effort to implement. Furthermore, this implementation of technology can be useful regardless of the type of maintenance performed, compared to interactive schematics which are mostly used in corrective maintenance practices.

5.3.6 Know where to start when having a fault

It is clear from the empirical findings that work orders with information regarding how specific faults were solved are saved, however, it is expressed that this information is not that easily accessible and also not used as much as it could be. There is therefore potential to utilize this information to guide technicians while performing troubleshooting activities and for them to know where to start when having a fault. Literature suggests that visualizing data including historical information about components and systems during troubleshooting processes can enable technicians to perform more efficiently [32]. Furthermore, the use of AI can help with analyzing historical maintenance data [10]. It is therefore reasonable to believe that collected historical work order data can be more easily utilized while incorporating AI to analyze, filter and find information quickly. Especially since it can be used even if a free text format was used during documentation. Incorporating AI into existing information systems will make filtering and searching

of historical work order data easier. Stored knowledge and previous interventions on a specific vehicle unit or information about how a fault on a component was fixed can be used as valuable insights for the technicians during troubleshooting activities. This implementation of technology could ease the challenge of knowing where to start when having a fault, as guidance from historical work orders can be accessed quickly, ultimately facilitating the challenge and time-consuming activity. Furthermore, since more guidance and help can be given, less experienced technicians can more easily troubleshoot and tackle faults.

Guidance can also be given using AR for remote support to help technicians know where to start when having a fault. This can especially serve as valuable if the right experience or competencies are not present in the depot. By having access to remote support via AR, information on how to go about a certain fault or recommendations according to previous experience can be given remotely. However, for this particular challenge, much guidance can be given by the knowledge stored in historical work order data. It is considered that using AI to analyze and filter the information is a quicker way of getting guidance on where to start when having a fault, furthermore, it is not dependent on an expert being available at all times. Therefore, using AI is considered to have a better trade-off, see Figure 14.



Figure 14: Evaluation matrix of digital tools and technology concepts relevant for the challenge of: Know where to start when having a fault.

5.3.7 Relying on information and help from experts

Communication and collaboration between actors in the maintenance process of rolling stock is expressed as rather complex and troublesome by respondents and

authors [2], [9], [13]. Moreover, the new ECM regulation clarifies that collaboration and communication between the functions and parties are and will be important [12], [16]. There are instances when the maintainer needs to interact with other actors with more competencies and experience in order to maintain the vehicle, but that interaction can be challenging and time-consuming. Digital tools and technologies, which can make communication and collaboration between parties easier, are therefore viewed as beneficial. As discussed, there are instances where technicians can get stuck in the maintenance process while having a lack of information or knowledge. Then knowledge and information need to be gathered elsewhere. The expertise and competencies needed are not always present in the depot, leading to technicians needing to communicate via email or phone to get support and guidance. This way of communicating is challenging and can take a lot of time. While incorporating AR technology, remote support and guidance can be achieved in a more descriptive and guiding manner. This can improve the communication between the parties resulting in more efficiently performed maintenance activities. Guidance and support can be achieved regardless of the location of the expert and there will be no need for extra traveling, hence being beneficial to this specific challenge, see Figure 15. Moreover, remote support and guidance provided by the vehicle manufacturer or suppliers can also be possible in order to troubleshoot and fix specific components that the technicians do not have as much experience or knowledge of. If the contractual agreements allow it, experts can use voice guidance, point to specific components, and give precise information and support on how to operate specific tasks, ensuring that the vehicle can be put into traffic faster. This way of communicating is direct with less time waiting for a response and it gives less room for communication obstacles since more contextual information is given. The remote support function utilized with AR technology can also be positive from an environmental point of view. Fewer components would be sent back and forth between the maintainer and the vehicle manufacturer or supplier if components can be fixed using remote support.

The challenge of relying on information and help from experts and other actors is hard to fully diminish as it is part of the maintenance process of the ecosystem. However, by using AR, the communication and collaboration between the actors can be improved, making the maintenance process more efficient. The use of AR for remote support can create opportunities for technicians to perform maintenance on components directly in the depot, and not waste time on sending components

to suppliers, or waiting for an expert to come to the depot to fix something. With this in mind, there is a business opportunity in the ecosystem of actors to provide a maintenance support service remotely using AR technology since the need for having better communication and collaboration between the maintainer and other actors is present and also the need to cover the lack of knowledge and competencies in depots. However, it needs to be a service that is accepted and can be given when needed, hence putting pressure on the availability of the service, so it can facilitate the challenge the technicians face. Therefore, the effort is considered high, but so are also the benefits, see Figure 15.



Figure 15: Evaluation matrix of digital tools and technology concepts relevant for the challenge of: Relying on information and help from experts.

6 Recommendation of Tools and Technology Concepts

The finalized recommendation for what digital tools and technologies to use and how to use them in order to facilitate the challenge and time-consuming activities identified is presented below.

- To facilitate the challenge and time-consuming activity of ***Navigating different information systems finding all relevant information***, the recommendation is to invest in tablets to easily access information and move away from using paper documentation. The next step is to go towards a more centralized information system where information is connected and reached quickly and no unnecessary time is spent on navigating different information systems. A Digital Model is recommended to be incorporated as part of this centralized information system as additional information in the form of technical description, inventory status, maintenance manuals and instructions, and schematics, can be linked to sub-systems and components of a 3D representation of the vehicle, giving a way of accessing and finding relevant information using a more graphical context.

- To facilitate the challenge and time-consuming activity of ***Locating material and equipment***, the recommendation is to incorporate solutions and practices of LEAN and 5S to create better structure and order in the depot and warehouse to easily find spare parts and not have equipment be misplaced.

- To facilitate the time-consuming activity of ***Needing vehicle in depot to retrieve logs***, the recommendation is to invest in eMaintenance technologies to have real-time data transfer to be able to analyze the logs earlier in the corrective maintenance process and have more time to establish the root cause of faults and also to optimize the scheduling and planning of preventive maintenance.

- To facilitate the challenge and time-consuming activity of ***Understanding instructions***, the recommendation is to invest in AR for instructions as it can facilitate the understanding and interpretation of instructions during

maintenance, making the process more efficient in terms of time but also the quality of work. Using this technology together with AR goggles the information can be displayed hands-free for a quicker interpretation. Furthermore, the knowledge and competencies needed to perform the maintenance will be lowered, making it easier to hire new technicians and tackle issues of lack of personnel. Steps to be taken are establishing what instructions are regarded to be harder to understand and incorporating these instructions in an AR application using animations, static 2D/3D models and text labels.

- To facilitate the challenge and time-consuming activity of ***Understanding placements of parts***, the recommendation is to invest in a Digital Model accessed via a tablet to have the ability to interact and investigate the configuration of the vehicle and connections between components in a 3D representation aiding with extra contextual information.

- To facilitate the challenge and time-consuming activity of ***Know where to start when having a fault***, the recommendation is to invest in AI to easier search, analyze and filter historical work order data to get guidance on how to tackle faults. By accessing the information on previous interventions performed and steps taken to solve a specific fault can stored knowledge be utilized and technicians can get help on where to start while having a fault.

- To facilitate the challenge and time-consuming activity of ***Relying on information and help from experts***, the recommendation is to utilize AR for remote support to minimize communication and collaboration difficulties when interacting with other actors in the maintenance process and procedures and create opportunities to fix faults faster and in the depot, ensuring vehicles are put back into traffic faster.

The recommendations displayed correlate to each of the specific challenges and time-consuming activities identified, but some results can be drawn on a general level on how digital tools and technologies can facilitate the work and procedures of technicians. Firstly, to make use of these digital tools and technologies mentioned supporting tools such as a tablet will be beneficial. Making tablets a standard tool for technicians to use opens up opportunities to easily find, access and visualize information close at hand. Secondly, incorporating more visual and graphical

context, in the form of 3D models, can help technicians both find, but also understand information. It is therefore recommended that incorporating more visual context can be useful in the maintenance procedures to facilitate the work of technicians. Lastly, there are also opportunities to connect actors and stakeholders, to create opportunities for better collaboration and communication, ensuring that maintenance can be performed more efficiently.

7 Conclusion

This chapter presents an answer to the research question. Furthermore, the limitations and recommended directions for future research are presented.

This thesis project presents new perspectives on how digital tools and technologies can facilitate the maintenance of railway vehicles. It is clear that technicians face many different challenges and time-consuming activities during maintenance procedures such as finding and accessing information, understanding information, and having successful support and guidance, but opportunities to facilitate these challenges using digital tools and technologies can be seen. The answer to the research questions *How can digital tools and technologies facilitate the maintenance of railway vehicles?* is that different tools and technologies can facilitate the maintenance of railway vehicles by visualizing and structuring information in a helpful manner, enabling the technicians to access and interpret information better. Furthermore, digital tools and technologies can facilitate better communication and collaboration between actors in the ecosystem by enabling sharing of data and knowledge.

7.1 Limitations and Future Research

The findings of this exploratory study could be seen as initial steps to understanding what challenges technicians performing maintenance on railway vehicles face and how digital tools and technologies can be used to facilitate the work and procedures of technicians. One limitation regarding the collected data is that only two depots have been visited and a handful of technicians working in depots have been interviewed. It can therefore be argued that perspectives and experiences can be limited and that there may be other challenges and time-consuming activities that remain unidentified. Furthermore, there may be other digital tools and technologies not presented in this study that can serve as beneficial in facilitating the work and procedures of technicians. Another interesting and also important finding is that the majority of respondents had a positive view on incorporating digital tools and technologies into the work and procedures, hence making the future research, development, and implementation of digital tools and technologies within the maintenance of railway vehicles more relevant in terms of acceptance. However, to fully understand the applicability of the recommended tools and technologies and their potential trade-off and use in the maintenance procedures of technicians,

further research and implementation are advised. Future research and the development of applications in the form of case studies where tools and technologies are implemented and tested together with technicians in their maintenance procedures is recommended to validate their use and that they would facilitate the work of technicians and minimize challenges and time-consuming activities present.

Another recommendation is to focus future research on different maintenance strategies performed in the depots since findings suggest that the work and procedures performed, as well as the potential use of tools and technologies, can be more or less beneficial depending on the type of maintenance. This is recommended to get deeper insights into what tools and technologies may be useful for a particular type of maintenance or specific depot.

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Appendices

Appendix A - Interview Information

Interview Information

About the project:
This interview is part of a master thesis project performed by Daniela Eichler, student at KTH in innovation management & product development. The thesis project is done together with WSP and Twindej.
The purpose of the thesis is to investigate what digital tools and technologies can facilitate the maintenance operations of rolling stock.
The purpose of the interview is to get an understanding of the maintenance process of rolling stock and what challenges there are for the technicians performing the maintenance.

The interview will be documented using notes and audio recording if it is accepted by the interviewee. The recording is only for transcription purposes and will not be published. The results from the interview will be analyzed and later presented in a report that will be published. Collected data and the interviewee will be anonymized.

Below are the areas and subjects that will be discussed in the interview:

Maintenance process:
- What is the current maintenance process
- Kinds of maintenance performed
- Routines
- Problems/Challenges with current maintenance process
- Challenging parts of the current maintenance process
- Tools and digital technologies
- Goals with the maintenance process

Documentation:
- Fault/Error reporting
- Documentation of the maintenance performed
- Information and manuals used

Collaboration:
- How is the collaboration between different departments/teams coordinated
- Collaboration and communication with stakeholders

I'm looking forward to meeting you and hearing about your work.
Best Regards
Daniela Eichler

Appendix B - Interview Guide

Interview Guide

About the project:
This interview is part of a master thesis project performed by Daniela Eichler, student at KTH in innovation management & product development. The thesis project is done together with WSP and Twindej.
The purpose of the thesis is to investigate what digital tools and technologies can facilitate the maintenance operations of rolling stock.
The purpose of the interview is to get an understanding of the maintenance process of rolling stock and what challenges there are for the technicians performing the maintenance.

Is it okay if I record this interview?
The purpose of the recording is to be able to transcribe the interview and later analyze the collected data. The data will be handled confidentially and will also be anonymized. The recording will later be deleted.

Initial questions:
- What is your name?
- What role and responsibilities do you have?
    - How long have you had this role?
- What education do you have?

Maintenance process:
- Can you walk me through a typical maintenance process
    - How do you know what maintenance to perform?
    - What is the routine for fault reporting?
    - Is it clear what type of information/documentation/manuals/instructions is needed and where to find it?
    - What are the challenges and difficulties with the current process?
    - What works well in the current process?

- What type of maintenance work is performed in the workshop?
    - The division between preventive and reactive maintenance?
    - Typical activities performed for each maintenance type and the time it takes to perform them?

- How is the maintenance work prioritized if multiple actions need to be taken?

- What are the most common problems you face in the maintenance process?
    - How do you handle these problems?

- Can you describe a specific challenging maintenance work/activity and how you solved that problem?

- Can you give me an example of when you needed to troubleshoot a complex fault?
    - How did you go about performing it?

- What digital tools, helpful aids and technologies do you use today when performing maintenance on rolling stock?
    - What needs do you have?
    - What tools and technologies would be helpful and why?

- How do you define good maintenance performed?
    - What does the contract between stakeholders look like?

Documentation:

- How does the documentation of maintenance work performed look like today?
    - What information system for documentation is used?

- What are you supposed to document/report to other stakeholders after maintenance work is performed?

- What documentation, manuals or instructions do you use while performing maintenance?

Collaboration:

- How is the maintenance work coordinated between the personnel and teams in the workshop?

- What is the work process for teams?
    - Always the same personnel working together or can personnel switch during a maintenance activity?

- How is the collaboration with the rolling stock operator?

- How is the collaboration with the rolling stock owner?

- How is the collaboration with the rolling stock manufacturer?
    - If needed, can you get in contact with the rolling stock manufacturer to better understand a subsystem of the vehicle?
    - If so, how often and how is it done?

Closing questions:

- Use your imagination, what does the future maintenance process of rolling stock look like?

- What opportunities do you see for implementing digital tools and helpful means to the maintenance work and procedures of rolling stock?
    - Any challenges?

- Do you have anything to add that I have missed regarding the topics we have discussed?

Appendix C - Interview Themes and Codes

Interview Theme	Code
Gather Logs from Vehicle to Perform Troubleshooting	Logs are valuable for troubleshooting
	Better if logs could be accessed quicker
Importance of the Right Competencies and Experience	Troubleshooting is much experience
	Lack of personnel in depot
	Lack of competencies
Handling of Material and Equipment	Lack of the right material to perform maintenance
	Tiring process to find what material to use and if the material is in stock
	Trouble locating material and equipment needed
Types of Information and Documentation used	Documents, instructions and manuals in pdf format and usually big files
	Instructions out-of-date
	Paper instructions still used
Work Order Documentation	Interventions performed are documented, but sometimes hard to understand
	Work order documentation is saved
	Using historical work order data to help with troubleshooting
	Filtering, searching and finding historical work order data is troublesome
Finding Information	Many different information systems and portals used on a daily basis
	Hard to search and filter information
	Information needed can be bundled within big documents
	Links in technical diagrams and schematics is helpful for technicians
Understanding Information and Instructions	Vehicle configuration can be hard to understand
	Hard to understand technical description, instructions and schematics, especially if less experienced technician
	Video instructions are valued to be easier to understand
	Visualization to help with understanding information
Interaction and Communication with Other Actors and Organizational Staff	Not always having all information needed from stakeholders and other actors
	Technicians are relying on other actors to fix components
	Fault reports can be hard to understand and interpret
	Communication and collaboration with stakeholders is challenging
View on Helpful Tools and Technologies in the Future	Potential in using VR for training and education
	Having guidance on steps while troubleshooting is desired
	Guidance given remotely could be potentially achieved with AR

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